<Global Standard Transport Labels and Waybills>

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Audience

The <Global Standard Transport Labels and Waybills> applies to the following function groups :

1. Network Operations
2. Customer Facing Solutions (ESS/ ESI)

Scope – Global/Regional/Country

Serial copy for CustomeXXX" only

Serial number:

Compliance

All locations where this activity is performed must comply with these guidelines.

If compliance cannot be achieved, local management must raise an exemption as per the Global SOP Change/Exemption Request Process.

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This document contains 82 numbered pages

1. Purpose

The purpose of this document is to supplement the current Global SOP procedures with finer detail to following core DHL Express Document Types:

- Transport Label
- eWaybill
- Loose BBX Transport Label
- ACS Transport Label

1.1. Out of Scope

- Manual House Waybill (HWB)
- Master Waybill (MAWB)

2. Objective

The objective of this Reference guide is to ensure all stakeholders:

- recognize and understand the business requirements on data and information on the Label
- (customers) recognize and understand the specifications, which must be adhered to and met, to be compliant with DHL EXPRESS' Label Standards, before accepted utilization is possible
- are able to identify issues and/or non-compliances with the Label and initiate mitigation actions

3. References

#	Document Name	Location/Link
R01	ISO/IEC 15459 Unique identifiers -- Part 1: Unique identifiers for transport units	International Standard
R02	ISO/IEC 16388 Automatic identification and data capture techniques -- Code 39 Barcode symbology specification	International Standard
R03	ISO/IEC 15417 Automatic identification and data capture techniques -- Code 128 Barcode symbology specification	International Standard
R04	ISO/IEC 15424 Information technology — Automatic identification and data capture techniques — Data Carrier Identifiers (including Symbology Identifiers)	International Standard
R05	ISO/IEC 15434 Automatic identification and data capture techniques -- Syntax for high-capacity ADC media	International Standard
R06	ISO 15394 Packaging — Barcode and two-dimensional symbols for shipping, transport and receiving labels	International Standard
R07	ISO/IEC 15416 Barcode print quality test specification — Linear symbols	International Standard
R08	ISO 3166-1, Codes for the representation of names of countries and their subdivisions — Part 1: Country codes	International Standard

R09	ISO/IEC 15418, Information technology — EAN/UCC Application Identifiers and Fact Data Identifiers and Maintenance	International Standard
R10	ISO/IEC 16022, Information technology — International symbology specification — Data Matrix	International Standard
R11	ISO/IEC 18004, Information technology — Automatic identification and data capture techniques — Barcode symbology — QR Code	International Standard
R12	Global SOP DHL Transport Label Specifications	Global SOP Archive
R13	Global SOP Label Certification Guideline	Global BCRS
R14	Global SOP Data Matrix 2D Specifications	Global BCRS
R15	Global SOP Customer-Facing QR-Code Specifications	Global BCRS

4. Definitions

4.1. Definitions

The following definitions are specific to this document:

Title	Description
DHL Transport Label	The **DHL Transport Label** is a label which is printed and attached to a piece and contains all handling-relevant information i.e. the one required to ship a piece from a Shipper to a Consignee. It includes piece/shipment identification, address information and DHL handling information. This generic title includes: • ESS Label (includes all information, mostly a Customer Produced Document as part of the Shipment Preparation Process, requiring no DHL re-labeling). It also includes specific sections for Customers to capture their bespoke information. • Network Label -> transport label without addresses and without customers' own segments • Routing Label -> small label containing DHL Handling Information only and used by DHL to complement information provided by customers.
eWaybill	A document type reflecting agreed "conditions of contract of transport" A **Waybill** is, by definition, **NOT** a label. This doc-type was historically called "Archive Doc" and later "Waybill Doc".
Multi-ply Waybill (Multiply Waybill)	Business term used as a reference for DHL's own Waybills doc-type (also named "Manual Waybill"). The term "Multiply Waybill" denotes both the empty and filled-in forms.
Reprinting	Term for a 1:1 replacement of a (damaged) label by a new one
Relabeling	Term for tagging a package with a new label with *additional* or changed supplementary information, e.g. Manual Sort Information.
License Plate	A Unique Identifier (code) assigned to a single transport unit by its issuer, in accordance with ISO/IEC 15459-1. It is the Leading Industry Standard Worldwide and acts as primary Piece Identifier (PID) standard for DHL Express. As such, DHL's equivalent business term is "**Piece Identifier**".
Shipment Identifier	Generic term for DHL Express identifiers at shipment level. Currently implemented with the 10 NUM format (9 digits, plus 10th digit calculated with MOD7). Other terms used within DHL are: • Air Waybill number (Air Express) • Waybill number (Air and Road Express) • Shipment number (Road Express) • Consignment number (Road Express, Freight), when the format matches the standard 10 NUM with MOD7 check-digit, defined above.
Reverse Video Format	A technique used in documents, whereby white characters are printed on a black background, as a visual aid to highlight an important section or piece of text.

4.2. Core Business Principles

DHL Labels (e.g. Transport Labels and eWaybills) are key Network documents as they enable and support efficient, physical handling of the shipment through to its delivery destination.

These Transport Label Specifications represent a Label Layout Specifications Document. They do NOT specify or define the corresponding data elements (defined in DHL EXPRESS data Model) but solely, the requirements that must be met by those (or systems) creating DHL-compliant labels.

4.3. Label Certification[1]

A DHL Label Certification is a pre-condition for a Transport Label to be released for production and use within the DHL EXPRESS Network. Five core elements are essential to that purpose:

- **Physical Label Layout (template)**
  - Customer labels must meet the Layout Requirements in order to optimize data readability and processing
  - Label Layouts are specified in this document and also cater for customer-specific requirements, through continuous updates based on customer feedback

- **Data / Information**
  - This element is vital because it covers the correct translation of the customer-driven (commercial) service order into specific DHL Handling Information paired with the latest global reference data
  - Due to its complexity and criticality, DHL strongly recommends this be NOT developed or managed by customers -> therefore it is not detailed in these specifications
  - Not all transport order data supplied by the customer must appear on the label. While a few data elements (e.g. product) must always be coded, only product features that may be relevant for sorting and delivery processes are shown on the label.
  - The data and information on the Transport Label does not replace the need for the transport order data. It enables however the piece to be physically moved through the network without the need of the corresponding data.
  - DHL offers, its customers, effective lookup solutions (e.g. GLSlabeldata) for automatically obtaining the correct set of label data corresponding to the ordered service
- **Rendering Solution**
  - DHL offers, its customers, effective rendering solutions (e.g. GLS Docrender)
  - DHL solutions are strongly recommended because they use the latest business logic, templates and data sources, which can't be guaranteed by other solutions (e.g. customer-owned) -> as a guiding principle, GLS is mandatory and ONLY GLS-rendered labels are acceptable
  - Business owners of Label Rendering / Printing Applications must ensure compliance to the valid Global SOP Standards is achieved

- **Printing Solution**
  - The minimal requirements regarding printing quality are specified in this document => they are to be met by customers prior to any usage in DHL's network
  - DHL's Label Certification Competence Center offers specialized support to customers in all development steps towards a compliant label

[1] For details on the Certification Process and/ or related Criteria, refer to the specific Global SOP Label Certification Guidelines

- **Label Certification Identifier**
  - A DHL Label Certification Identifier is a pre-condition for a Transport Label to be released for production and use within the DHL EXPRESS network
  - The use of DHL's solutions (e.g. GLS w. lookup & rendering features) automatically ensures the Label Certification Criteria is met

4.4. Documentation-related Principles

- A **mandatory**-marked field implies that the content is **ALWAYS mandatory**. This implies that its absence is Not acceptable because it impacts the quality of the label data/ info and thus of the associated shipment/ piece handling processes.
  - A **Conditional**-marked field implies that the content is mandatory in some cases (e.g. for a specific product) and optional in other cases. Further details accompany the specific fields
  - An *Optional*-marked field indicates that the content is not required, but may be added if data is available and brings value. Optional does not always imply that the field may remain empty. Further details accompany the specific fields and may require that for example they have to be populated with at least zeroes.

4.4.1.Legend

CAS *Information valid solely for Customer Facing Solutions/ Systems (e.g. EDI and DHL ESS / ESI solutions)*

Cross-references are marked with this symbol. They are provided to indicate related chapters or to refer to associated information.

Product-specific information is marked with this symbol

This symbol indicates country- or region-specific legal requirements

Important remarks are marked with a red exclamation mark.

5. Transport Label – Segment Specifications

Because several Data Elements and Fields are common to both the DHL ***Transport Label*** and the DHL e***Waybill***, which build the core set of the Global Label Standards, the listing and definition of the respective components will be sub-divided into Label- and Order-related.

As a note, a label consists of sections, which may further contain one or more segments.

The numbered sections of the DHL Transport Label as shown below refer to the following paragraphs where the sections are described in more detail. The dimensions of the sections are described in the technical specification.

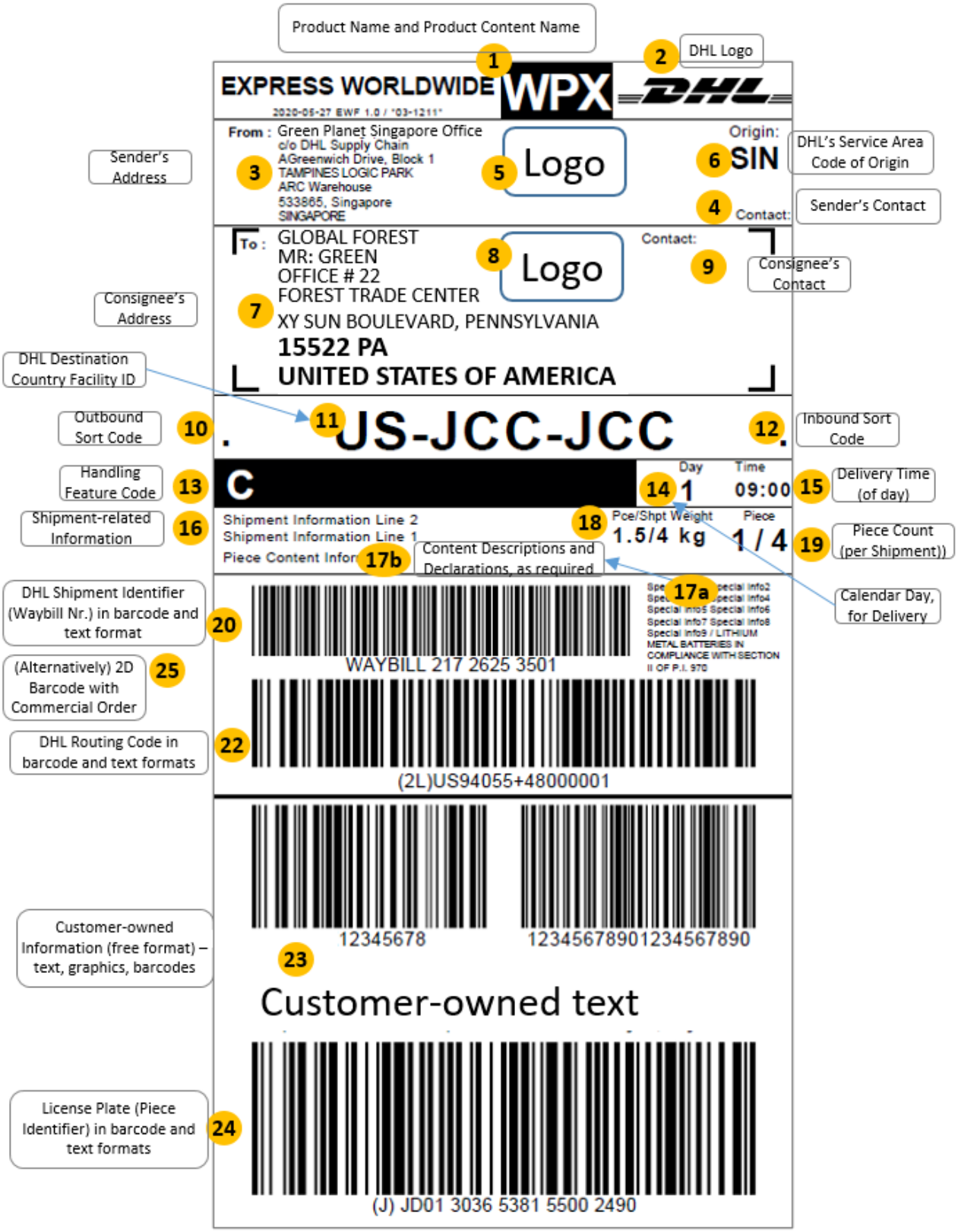
This chapter defines the contents of the sections and segments and their character (i.e. if they are mandatory, conditional or optional).

- **Mandatory**: means it must always appear independent of the scenario. If mandatory information is not available, the whole label or a particular label section/segment should not be printed. (E.g.: no DHL Transport Label must be printed without the Piece Identifier barcode)
- **Conditional**: means it may be mandatory in some cases and optional in other cases. (e.g. depending on product or country of origin)
- **Optional**: means that the label field content is not mandatory, it does not have to be displayed on the label. (E.g.: the customer logo)

Note: For the routing information, the optional content information does not always indicate that the dedicated section or segment field remains empty/blank when printing the label. For instance, if the optional delivery date or time features are not chosen, optional implies that zeroes are displayed in the routing barcode.

5.1. Layout overview [2]– DHL Transport Label (all segments)

[2] This page offers a structured overview of the Global SOP Standard Transport Label. Sub-sequent sections explain in detail the specifications of each individually numbered segment.


Product Name and Product Content Name
1
2
DHL Logo
EXPRESS WORLDWIDE
WPX
2020-05-27 EWF 1.0 / "03-1211"
From : Green Planet Singapore Office
c/o DHL Supply Chain
AGreenwich Drive, Block 1
TAMPINES LOGIC PARK
ARC Warehouse
533865, Singapore
SINGAPORE
Sender's Address
3
5
Logo
Origin:
6
SIN
DHL's Service Area Code of Origin
4
Contact:
Sender's Contact
To : GLOBAL FOREST
MR: GREEN
OFFICE # 22
FOREST TRADE CENTER
XY SUN BOULEVARD, PENNSYLVANIA
15522 PA
UNITED STATES OF AMERICA
8
Logo
Contact:
9
Consignee's Contact
Consignee's Address
7
DHL Destination Country Facility ID
Outbound Sort Code
10
11
US-JCC-JCC
12
Inbound Sort Code
Handling Feature Code
13
C
Day
Time
14
1
09:00
15
Delivery Time (of day)
Shipment-related Information
16
Shipment Information Line 2
Shipment Information Line 1
Piece Content Infor
17b
Content Descriptions and Declarations, as required
18
Pce/Shpt Weight
1.5/4 kg
Piece
1 / 4
19
Piece Count (per Shipment))
Calendar Day, for Delivery
DHL Shipment Identifier (Waybill Nr.) in barcode and text format
20
17a
Special info2
Special info4
Special info5 Special info6
Special info7 Special info8
Special info9 / LITHIUM METAL BATTERIES IN COMPLIANCE WITH SECTION II OF P.I. 970
WAYBILL 217 2625 3501
(Alternatively) 2D Barcode with Commercial Order
25
DHL Routing Code in barcode and text formats
22
(2L)US94055+48000001
Customer-owned Information (free format) – text, graphics, barcodes
12345678
12345678901234567890
23
Customer-owned text
License Plate (Piece Identifier) in barcode and text formats
24
(J) JD01 3036 5381 5500 2490

5.1.1. Standard Label Sample with operational data

5.1.2.Standard 2D-enabled Label Sample

5.2. Product Name, Product Content Code and Version Name

1 This section contains the name of the Product used (e.g. EXPRESS WORLDWIDE) and the three-letter Product Content Code (e.g. WPX or EPL), in accordance with the DHL Express global product portfolio.

This Reference Data is maintained in DHL's Global Reference Databases.

5.2.1. Product Name (Conditional)

The Global Product Name officially specifies the product under which the underlying transport service has been sold to the customer and is to be printed here with the associated format "Product Short Name". This is a text string in upper-case letters, up to 20 chars long (reason for the Short Name). It may contain the full set of 7 bit standard chars, including white space, upper/lower case chars and symbols such as a colon (":").

The conditional qualifier is applicable to the "Compact Label", which use requires a formal approval (see below).

5.2.1.1. Highly-Compact Label[3]

There are scenarios when, due to customer requirements for extra info or elements (e.g. additional barcodes) on the label (e.g. in the customer-owned segment), there is the need to maximize the compactness of the info on the remaining Label segments in order to free up space elsewhere.

In such cases and upon special approval by DHL EXPRESS, the Product Name may be omitted.

5.2.2. Product Content Code (mandatory)

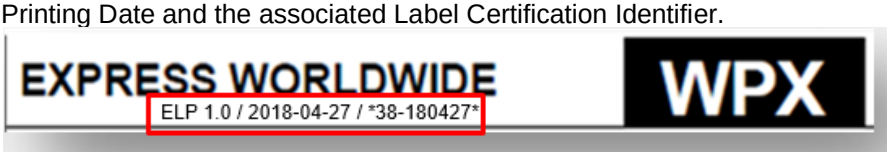
This is DHL's internal code for the product under which a certain transport order will be processed. It always consists of 3 upper-case characters. Different products may have the same Product Content Code

For declarable shipments, the Product Content Code must be printed in reverse video [4]format (see image below, in 5.2.3).

Product Name and Product Content Code can be determined based on the required (i) Service level (e.g. pre-9:00) and (ii) DOC or NON-DOC shipment.

5.2.3. Printing Application, Date and Label Certification Identifier (mandatory)

The information herewith required has a mandatory char and aims at identifying both the label Printing Application, the Printing Date and the associated Label Certification Identifier.



There is no binding format for the Printing Application (e.g. "Express Logistics Platform - ELP") but there is one for its version: X.XX.XXX. It shall be located underneath the Product Short Name.

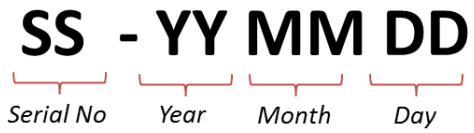
[3] Refer to section 8.1.6 for further specific details on the Highly-Compact Label

[4] Refer to Definitions section

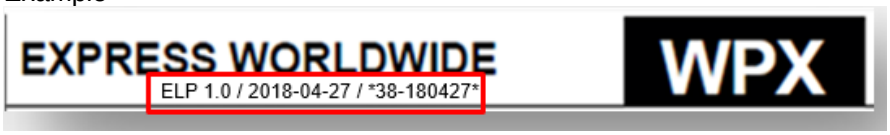
5.2.3.1. Permanent Label Certification Identifier (**mandatory**)

All Label Certification Identifiers are managed centrally, by the GCoCL team and follow a sequential scheme. There is a differentiation between **permanently** and **temporary** approved labels which still require some rework and need to be reviewed after six month of approval.

Permanent Label Certification Identifier Convention:

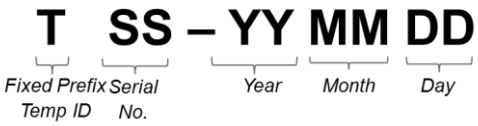


Example



5.2.3.2. Temporary Label Certification Identifier

Temporary Label Certification Identifier Convention:
A **temporary certification** can be used for a limited period but must be reviewed after 6 months to fully adhere to the DHL Express standards. It will follow a different naming convention starting with a fix suffix followed by a serial number and date. Once fully compliant, the temporary ID will be changed to a permanent ID



The font size for the Certification Identifiers is of a 1.5mm font size or 5 points.

5.2.4. Label Printing Date (**mandatory**)

The Label Printing Date, in accordance with the ISO 8601 standard, follows (sequence) the identification of the Rendering Application and has the format yyyy-mm-dd, [5] where "YYYY" refers to the year, "mm" to the month and "dd" to the date.

5.2.4.1. Full Label Segment

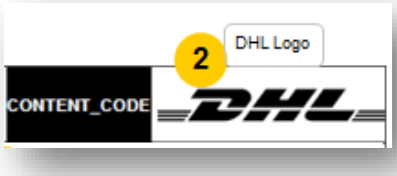
The full format of this segment is, thus:

"[PRINTING APPLICATION]"/"[PRINTING DATE]"/"* [CERTIFICATION IDENTIFIER]*

[5] Based on GLS server date and time.

5.3. DHL Logo (mandatory)

This section contains the carrier logo and is placed in the upper right corner of the label. Directly below the logo, associated information can be printed (e.g. a particular DHL Country Organization).



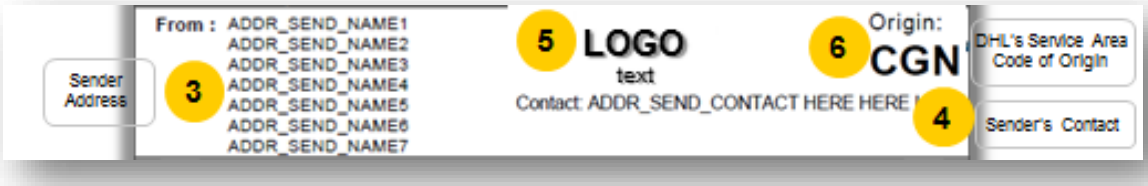
For transport labels, the black & white version of the logo is sufficient. When using labels with a pre-printed logo, the official DP DHL brand color specifications for the logo have to be followed.

Any DHL-owned customer automation system must be able to print the logo as an image. Only if it is not possible to insert images into the label, may the equivalent text be printed in plain text instead.

Always use the correct logo layout and specifications, as per the official DP-DHL Standard Brand and Design Guidelines.

5.4. Ship 'From' Address and Customer Logo

5.4.1. 'From' Address



The "Ship From" section refers to the Consignor's Address. It serves the purpose of supporting identification by the carrier (e.g. for Pickup) as well as the Consignee. It does not necessarily reflect the return address in case of failed delivery. If approved by DHL, a deviating address may be printed here.

The word "From" is to be printed in the upper left corner of this section. It may be written in a local language, however for international shipments the English word "From" must appear as well. In this case only English, or both languages (English and local language) may be used, separated by a "/" symbol.

The data Address elements and their definition are based upon the DHL Corporate Data Model, which is an integrated view of the data produced and consumed across DHL's organization. It unites, formalizes and represents things important to DHL, as well as the rules governing them.

The maximum number of lines is restricted by DHL's internal data flow and the maximum height of the block. Each line contains a maximum number of characters determined by the country data configured in GREF.

Additional information (e.g. floor number, county) is possible but the overall number of lines must not exceed 7, for consistency with DHL's data structure.

For international shipments, the Country Name shall be written in its own line.

For domestic pieces the Country Name may be printed in local language. For all cross-border pieces the Country Name has to be printed in English (may be followed by local translation). The Country Name is optional for domestic shipments BUT mandatory for cross-border shipments if ISO 3166 code is missing)

The Country Code follows the ISO 3166 two-character country code standard and must be used whenever available as a quick reference for a number of DHL services.

5.4.1.1. Highly-Compact Label

For these cases and upon formal DHL EXPRESS approval, the number of Sender Address lines can be limited to 5 (from the allowed 7 lines) with the functional advantage that the font size can be kept high with the inherent advantages in terms of readability. This is achievable with the following reference structure:

1. Address line 1: Receiver Company / Name
2. Address line 2: Building Name or equivalent
3. Address line 3: Street Name, Street Number
4. Address line 4: City Name, Postcode or equivalent for other Postal Location Formats (e.g. Suburb, County, State)
5. Country Code / Country Name

5.5. Customer Logo

The relative small font of the sender address offers space to the right of the segment. This is reserved for an optional Customer Logo or alternatively for a DHL Initiative Logo (e.g. GoGreen) if customer agrees or asks for it.

5.6. Consignor's Contact (Conditional)

Shipper contact details are still critical operational information and will be captured and stored, in accordance with GDPR, as part of the shipment preparation process.

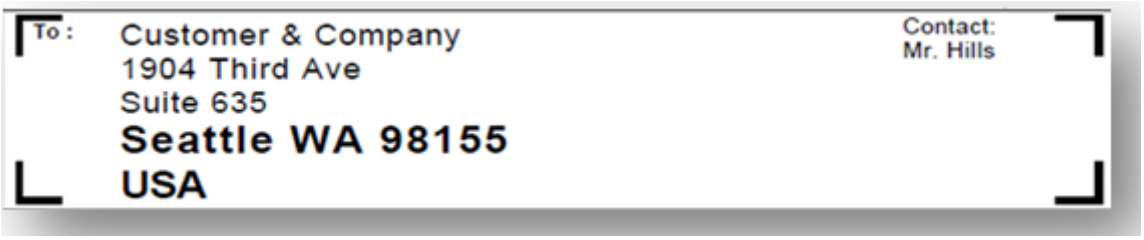
As part of DHL's ongoing process and systems improvements, this information is, however, no longer required to be printed on the Transport Label. It is made available electronically to our relevant employees and operators to perform their tasks and deliver the service expected of our customers.

Note: in exceptional circumstances (e.g. for compliance with international regulations like the ICAO or IATA for goods with UN Code UN3373) and only upon formal approval, specific contact details may have to be printed on the Transport Label.

5.7. Service Area Code of Origin (mandatory)

DHL Express needs the Service Area Code of Origin printed in this segment. This is a 3-digit, upper-case letter, of a locally defined and globally recognized area where DHL services are provided. This data is maintained in DHL Global Reference Databases.

5.8. 'Ship To' Address (mandatory)



Each piece requires a Ship To address. This is the agreed Delivery Point.

The word "To" may be written in local language, however for export shipments the English word "To" must appear as well. Alternatively, both languages (English and domestic) may be used, separated with a "/".

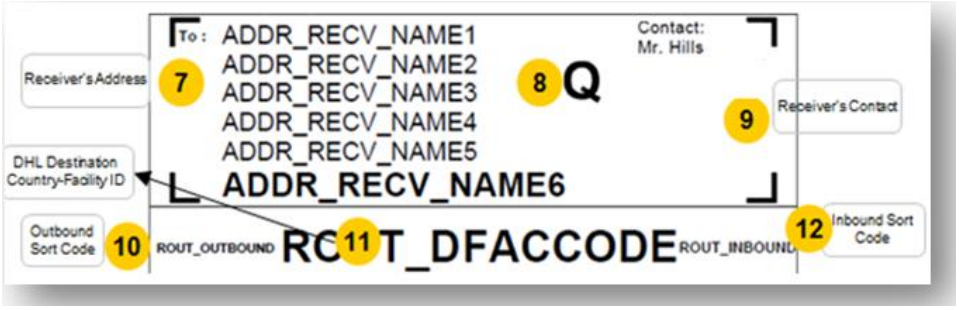
The Ship To address should be emphasized by adding hooks, i.e. thicker corners. The recommended size of the hooks is 4 x 4mm and the line thickness is > 0.5mm. The area contained by the hooks shall include the contact information.

For the structure of the Ship To address, refer to section 3.4.1 Ship From address

Any customer automation application has to offer a minimum of 5 lines for the Consignee address.

Further information such as floor number, county etc is possible but the overall number of lines must not exceed 7, for consistency with DHL's data structure.

Delivery Country and the applicable Postal Location Format Elements (e.g. City and Postal Code) have to be printed in bigger font. In case of lack of space, the two lines can be joined to one line.



7 **Delivery Country** and the applicable **Postal Location Format Elements** (e.g. City and Postal Code) have to be printed in bigger font. In case of lack of space, the two lines can be joined to one line.

8 This area of the label is reserved for an optional Customer Logo, represented by Q in this diagram, or alternatively for a DHL Initiative Logo (e.g. GoGreen) if customer agrees or asks for it.

9 Consignee's Shipper contact details are still critical operational information and will be captured and stored, in accordance with GDPR, as part of the shipment preparation.

As part of DHL's ongoing process and systems improvements, this information is, however, no longer required to be printed on the Transport Label. It is made available electronically to our relevant employess and operators to perform their tasks and deliver the service expected of our customers.

Note: in exceptional circumstances (e.g. for compliance with international regulations like the ICAO or IATA for goods with UN Code UN3373) and only upon formal approval, specific contact details may have to be printed on the Transport Label.

5.8.1. Highly-Compact Version

In analogy to the Consignor's Address segment, there may be the need to maximize the compactness of the info on the remaining Label segments in order to free up space elsewhere.

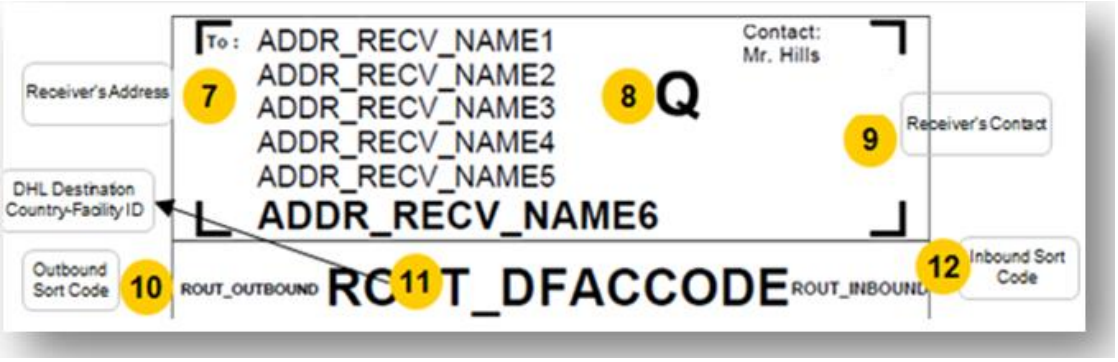
For these cases and upon formal approval, the number of Consignor Address lines can be reduced to 5 with the functional advantage that the font size can be kept high with the inherent advantages in terms of readability. This is achievable with the following reference structure:

1. Address line 1: Receiver Company / Name
2. Address line 2: Building Name or equivalent
3. Address line 3: Street Name, Street Number
4. Address line 4: City Name, Postcode or equivalent for other Postal Location Formats (e.g. Suburb, County, State)
5. Country Code / Country Name

5.9. Manual Sorting Section

DHL's Manual Sort Information consists of the following three segments, from left to right:

- Outbound Sort Code -> element 10
- Destination Facility Code -> element 11
- Inbound Sort Code -> element 12



5.9.1. Outbound Sort Code (Conditional)

Each DHL Service Area can define an Outbound Sort Code per Destination Facility Code. Sort Codes consist of up to four alphanumeric chars (numerals or capital letters).

DHL Outbound Sort Codes are **mandatory** **when defined** (which is not the case for all outbound facilities) and must be printed on the label to support manual sorts. The corresponding information is available in DHL's Global Reference Database (GREF) which must be referred to for the latest valid values.

In cases where no Outbound Sort Code is defined by DHL Express in the country of origin, this field is to be populated with a "." This is equivalent to the statement that "." is only used if an effective attempt to look a sort code up returned no value.

Whenever no attempt was made to determine the Outbound Sort Code, then this should be left blank. "." This will also indicate to DHL that the label needs to be re-printed.

5.9.2. Destination Facility Code

This segment contains the complete DHL Facility Code, of the DHL Facility which will manage the Delivery and consists of:

- Country Code (two-digit: capital letters according to ISO 3166)
- hyphen
- DHL Service Area Code (three-digit; capital letters or numerals)
- hyphen
- DHL Facility Code (three-digit; capital letters or numerals)

The Destination Facility Code depends on destination country, postal location type (e.g. post code) and product. This data is kept constantly updated in DHL's Global Reference databases.

In cases where a DHL Facility Code has not been specified or made available for a certain destination or an Address cannot be matched with Geo Location Coordinates in GREF, only Country Code and Service Area Code are to be printed (separated by a hyphen). Example: ***SG-SIN*** *for Service Area Singapore*

5.9.3. Inbound Sort Code (Conditional)

DHL Inbound Sort Codes are **mandatory** and must be printed on the label when defined (not the case for all inbound facilities).

12 The Inbound Sort Code consists of up to 4 alphanumeric chars (numerals or capital letters). It depends on destination country, Postcode and product and can serve as either:

- Consolidator: various facilities having the same Inbound Sort Code
- Separator: A facility sub-divided in areas, based on Postal Code and/or product

DHL may not require the Inbound Sort Code for all destinations, in which cases this field is to be populated with a “.”.This is equivalent to the statement that "." is only used if an effective attempt to look a sort code up returned no value.

Whenever no attempt was made to determine the Inbound Sort Code, then this should be left blank. "." This will also indicate to DHL that the label needs to be re-printed.

5.10. Product Features

5.10.1. Handling-relevant Feature Codes - HFC (Conditional)

Product Features encompass Handling and Date/Time Delivery features, which are shown on the label to support DHL internal operational procedures.

Only product features which are **handling-relevant** (i.e. impact the sorting and/or delivery processes) are shown on the label (and printed left aligned). The Handling Feature Codes (e.g. “DTP”), translated from the chosen Product and Service Codes (e.g. “DD”) are printed in reverse video format in this segment.

When defined through the Service Codes, **HFCs are mandatory** and must be printed as per the Global Standard.

The translation from commercial products and services to handling codes is based on DHL’s reference data (e.g. tools provided by DHL Express to our customers offer this capability).

Product features that are not handling-relevant appear only on the accompanying documentation, such as the electronic information (EDI), the pickup- and delivery lists.

5.10.2. Delivery Date (conditional)

This field shows the delivery date feature (to be coded mandatorily in the routing barcode - – refer to section 3.12.1).

The Delivery Date/Time information has a header, consisting of the words “Day” and “Time”, which may be written in a local language. However, for international shipments, the English words “Day” and “Time” must appear as well.

The separator “-” between Date and Time field is required whenever at least one of both features is chosen.

The specified Day of the Month is to be represented in 2 digit numeric format with leading zeroes.

This field may also contain non-numeric values, representing delivery instructions that cannot be combined with a certain delivery date.

Examples:

Delivery Date codes	Product	(In Routing Barcode)	Label Code
No fixed date delivery	All	00	(none)
Fixed day delivery (day of the month)	…	01 … 31	01 … 31
Delivery on appointment	…	51	A
Saturday only delivery	…	50	S
Hold at depot (no delivery)	…	52	H
…	...	...	...

The translation from commercial products and services to handling codes is based on DHL's reference data (e.g. tools provided by DHL Express to our customers offer this capability).

5.10.3. Delivery Time (conditional)

If a time window is chosen, the code corresponding to the time window is shown (to be always coded mandatorily in the routing barcode – refer to section 3.12.1).

Examples:

Time Feature Text	(In Routing Barcode)	Handling Information
No fixed delivery time	0	(none)
Pre 09:00	1	"X09"
Pre 10:30	3	"X10"
Pre 12:00	2	"X12"
…	…	…

The translation from commercial products and services to handling codes is based on DHL's reference data (e.g. tools provided by DHL Express to our customers offer this capability).

5.10.4. Handling-relevant Service Logos

5.10.4.1. Medical EXPRESS Logo

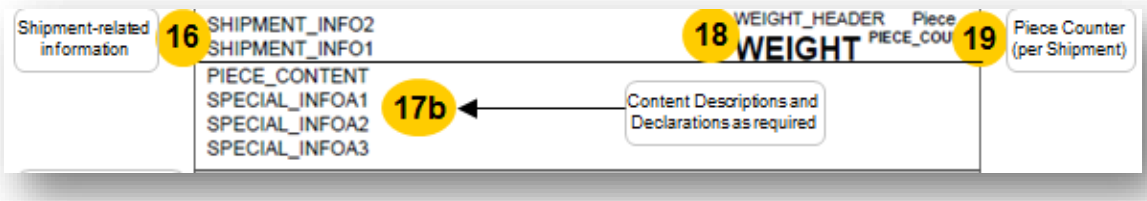
In addition to Handling Feature Codes, there are cases when extra information is judged valuable in ensuring the optimal process management quality.

Some of those cases are Medical Express Shipments, which may contain life-saving commodities (e.g. vaccines, medicines) and/or are temperature-sensitive and, as such, need to be identified, processed and delivered with the highest possible priority and accuracy.
The addition of a Medical-related symbol (logo) to the area adjacent (above) to the scannable Piece Identifier barcodes (refer to image below) will help keep it in full sight of the operations team at point of processing.

This new logo is unique6 to Medical Express shipments so that it continues to remain a distinct and the powerful signal that priority is required. The chosen logo is the heart shape with a pulse line transitioning through it (refer to image below).

A min area of 15*15 mm2 is specified for the logo and must be complied with, while the original proportions must be kept. The location of the Medical EXPRESS Logo shall not impact or overlap the necessary quiet zones of the adjacent barcodes.

5.11. Standard Shipment information



The Shipment Information section contains the following information elements:

5.11.1. Piece / Shipment Reference (optional)

[6] The Medical EXPRESS Logo is specified in compliance with the applicable Corporate Branding guidelines and is available, in the form of an approved graphical file, for operational use.

A customer may refer to a package using specific identifier(s) (e.g. reference codes, which consist of sets of alphanumeric chars). A Shipper's Reference is one type of the more generic Shipment References, which is under customer's responsibility to manage.

Shipment References may be printed on the label, ideally in cases where DHL has received those references with the transport order. All formats supported by the solution agreed with DHL can be printed on the label as well.

Piece and Shipment References are usually qualified by specific Reference Type Codes (ex-Reference Qualifiers), which give them a specific meaning. Refer to DHL EXPRESS Piece/Shipment Reference Standard.

5.11.2. Piece / Shipment Weight (conditional)

The weight of all single pieces ("Shipment Weight" = weight of all pieces building up a shipment) and of the corresponding single(s) piece are to be printed here.

The lack of Shipment Weight may lead to issues in Customs processing and is therefore ***mandatory*** ***for at least one package of each dutiable shipment****.*

As a minimum requirement, the label provided with the last piece must contain both the sum of the weight of all pieces belonging to the shipment (legacy term "Shipment Weight") and its corresponding single piece weight.

Example:8/25 for the last piece label of a shipment, where "8" corresponds to the single piece weight and "25" corresponds to the sum of all single pieces building up the associated shipment. The initial piece labels would have the value "7/.", where the "." indicates that the total weight is not yet known.

For single-piece shipments, the weight needs to be printed only once (e.g. 13.4 kg instead of 13.4/13.4 kg)

The ***Transport Label*** *always has to show the* ***Dead Weight****, NOT a calculated volumetric weight, should it be available at the time of printing the label*

5.11.2.1. "Shipment Weight" – Shipper Declared vs DHL Actual

Weight, as provided by the customer (e.g. "Original Shipper Declared Weight" through EDI / 2D Code / Manual Waybill), has to be printed on the label if the printing occurs prior to DHL Reweigh processing, in this field.

When no weight was available at the time of printing the label, a "." shall be printed to indicate that fact.

Customs recommends that, for easier customs processing, the full shipment weight i.e. the total weight of the pieces making up the shipment, be printed on each package. This should be done where this information is available when printing the label.

Note: the sum of all pieces weight is not always known at the time of printing a label. For this reason and in such circumstances, it is acceptable that only the last piece label contains the set [Piece Weight] / [Total Piece Weight].

Any customer automation application has to ensure that at least the last package of a shipment contains the shipment weight. Wherever the shipment weight is known prior to printing the first label, it must be printed on all labels.

5.11.2.2. Piece Weight

Piece Weight is **mandatory** for Remote Pickup shipments and must be printed whenever available for multi-piece shipments.

Piece weight may be legally required in some countries, for health and security reasons.

5.11.2.3. Printing Structure

a) Measuring Unit
The weight can be stated in kg or in lbs, alternatively. The measuring unit has to be printed following the weight value. In case of combined piece/shipment weight it is only required behind the shipment weight. The measuring unit for both weights must be the same in all cases.

b) Value
The weight value must be a valid number, with no more than one decimal digit

c) Header Text
- The header text for a piece weight is “Piece Weight”
- The header text for a shipment weight is “Shipment Weight”
- Where both piece and shipment weights are printed, the header text is to be contracted to “Piece/Shipment Weight”.

Abbreviations such as “Shpt” for Shipment or “Pce” for Piece are allowed. For domestic shipments, the English terms may be translated into local language.

5.11.2.4. Examples

a) Printing both piece and shipment weight	Piece / Shipment Weight: 5.5 / 33.8 [kg]
b) Printing shipment weight only [7]	Shipment Weight: 33.8 [kg]
c) Printing piece weight only	Piece Weight: 8.0 [lb]
d) Printing weight for one-piece shipments	Piece / Shpt Weight: 2.7 [kg]

5.11.3. Relative and Total Number of Pieces in Shipment (conditional)

DHL needs the exact number of pieces to be transported and billed. This entry is predominantly **mandatory**, as it reduces the risk of delays in customs processing.

The overall criticality of the availability of the piece count varies between “very high” for non-doc and “moderate” for “doc” shipments. The piece count should be suppressed if its correctness cannot be guaranteed.

[7] This legacy term refers to the sum of all pieces weight. The Waybill Document (see section 6.10) must display, at a minimum, the shipment weight.

The following rules apply:

1. Any single-piece shipment has to carry the string "1/1", where a piece counter is separated with a slash « / » from the total number of pieces.
2. As a minimum, the label provided with the last piece must contain both the total number of pieces belonging to the shipment and its final single count (e.g. 12/12)
3. DHL's approval is explicitly required for a permanent suppression, on account level, of the relative and/or total number of pieces.
   a. Any missing number is then to be replaced by a "." (dot)
   b. Example: "3/." Indicates the third piece of a shipment where the total number of pieces is not known at the time of printing the label
   c. As a summary:
      i. "X/." is the syntax for all labels as long as the total planned number of pieces is not known
      ii. "X/Y" is the syntax for all labels as soon as the total planned number of pieces is known
      iii. "Y/Y" is the syntax for the last piece label of a shipment

5.11.4. Piece Weight Symbols

Customers, Regulators or Standard Bodies sometimes require additional information to be added to Transport Labels. This is, for example, the case with Germany, where a new law was passed on 01 July 2024 requiring that specific icons or symbols be applied to warn users about piece weight being above specific thresholds. It is necessary to consider that similar rules may be applied in the future by other countries or handling processes.

5.11.4.1. Germany (mandatory) - background

A federal law imposes that, besides a clear indication of each piece's weight on the Transport Label, additional specific icons or symbols be displayed, when certain conditions are met. This has the purpose of ensuring:

- users become aware of a piece weight being above specific thresholds
- pieces are handled safely, potentially, through the use of appropriate equipment

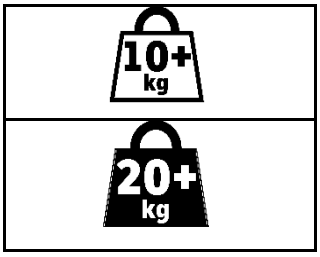
This rule applies to all deliveries with a final destination in Germany (DE), whether they are domestic or international shipments. In the future, similar rules might also cover pickups and other handling processes. The rule does not apply to shipment transit via Germany.

5.11.4.1.1. Scope

The changes will be implemented across all Transport Label templates (e.g. ECOM, NETW and DUO).

5.11.4.1.2. Weight Symbols

Weight Symbols



5.11.4.1.3. Business Rules

The following business logic and rules apply:

1) The Weight Symbols shall be applied for all shipments delivered in Germany (DE), on all the respective Transport Labels

2) Only “PIECE_WEIGHT” (Customer-Declared Piece Weight) will be used to determine the weight symbol rendering; “SHIPMENT_WEIGHT” will not be considered

   a) If PIECE_WEIGHT is less than 10 kg or 22 lb, no Weight Symbol will be displayed

   b) If PIECE_WEIGHT is equal to or greater than 10 kg or 22 lb but less than 20 kg or 44 lb, a "10+ kg" symbol shall be displayed on the Transport Label

   c) If PIECE_WEIGHT is equal or greater than 20 kg or 44 lb, a "20+ kg" symbol shall be displayed on the Transport Label

3) The Weight Symbols shall be applied irrespective of the original unit of measurement provided by the Customer (e.g. kilogram (kg) or pound (lb) in the WEIGHT_UOM message field

Summary

No	Piece Weight Specification	Weight Range	Weight Icon Printed
1	Less than 10 KG or 22 LB	0.100 to 9.999 KG 0.100 to 21.999 LB	No Icon printed
2	Equal to or greater than 10 KG or 22 LB but less than 20 KG or 44 LB	10.000 to 19.999 KG 22.000 to 43.999 LB	
3	Equal to or greater than 20 KG or 44 LB	20.000 KG or 44.000 LB and above	

5.11.4.2. Standard Location of Weight Symbols

The new Weight Symbol shall be located in the current segment containing the Piece and Shipment Weights, on their left side.

To ensure that the weight icon is easily visible in all distinct templates and does not overlap with other shipment fields, fine adjustments and variations of the Symbol locations are acceptable.

Implementation of deviations from the Global SOP Standard Transport Label and respective Weight Symbol location require pre-approval from Global SOP or from the Global Competence Center for Labels and Barcodes.

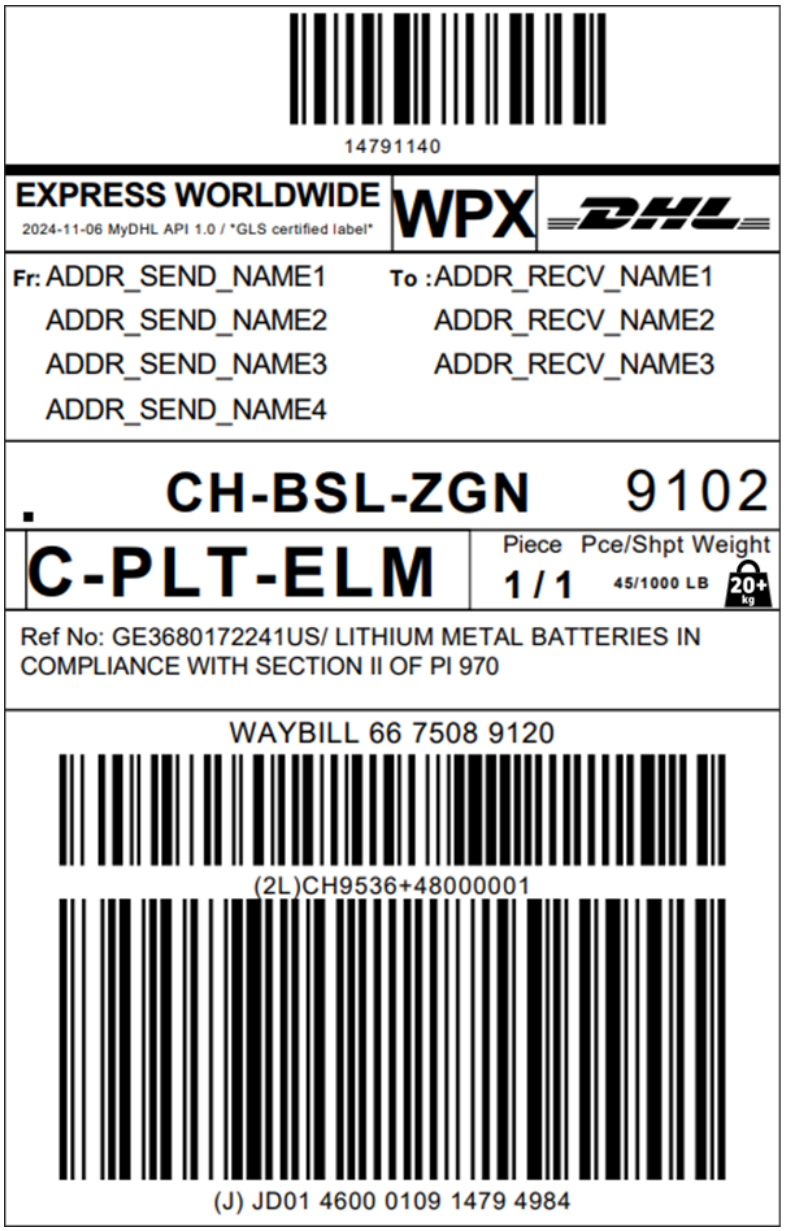
Next, two samples showcasing the application of the Weight Symbols on a standard 8x4 label and a smaller 6x4 label:

8x4 Label Size

6x4 Label Size

5.11.4.2.1. Special[8] Location of Weight Symbols

ECOM26_711_001
Due to limited available space on the ECOM26_711_001 template, the new Weight Symbol will be printed to the right of the Pce/Shpt Weight segment:

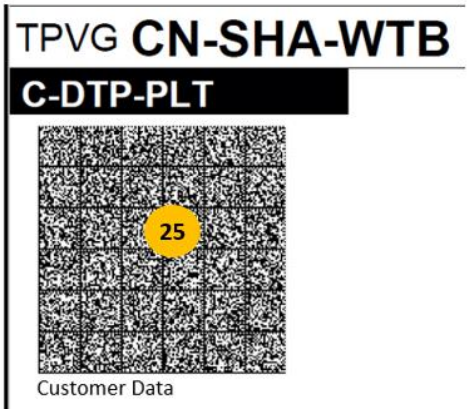


[8] Implementation of deviations from the Global SOP Standard Transport Label and respective Weight Symbol location require pre-approval from Global SOP or from the Global Competence Center for Labels and Barcodes.

5.12. 2D Data Matrix – Customer Data

Data Matrix is a powerful two-dimensional matrix symbology selected for encoding operational data to be used in the operational landscape. It is made up of square modules arranged in a square or rectangle shaped pattern.

For further details on both the symbology and key 2D code parameters, refer to section 9.3. " Bi-Dimensional Barcode Symbologies".



[9]

The method for storing data in 2D Codes is specified in the ISO/IEC 15434 (specification and syntax for automatic identification and data capture techniques commonly used in 2D barcodes such as Data Matrix.). This ISO/IEC standard features the possible use of multiple segments with different contents and might be seen as the file system of the code.

The following segments are available and they are always in the sequence below. Only the first is mandatory but both the public and open segments are in systematical use by DHL EXPRESS:

- **Public segment**: standardized package data shared between customer, service providers and DHL
- **Open segment**: open segment in pseudo EDIFACT DHL data structure shared between service providers and DHL
- **Customer segment**: segments with customer data not used by DHL

5.12.1. Customer Data

The 2D symbol under use contains encoded operational data, which is obtained through the concatenation of the above listed segments' data.

The exhaustive list of encoded shipment data elements can be referenced through the Global SOP "Data Matrix 2D Specifications".

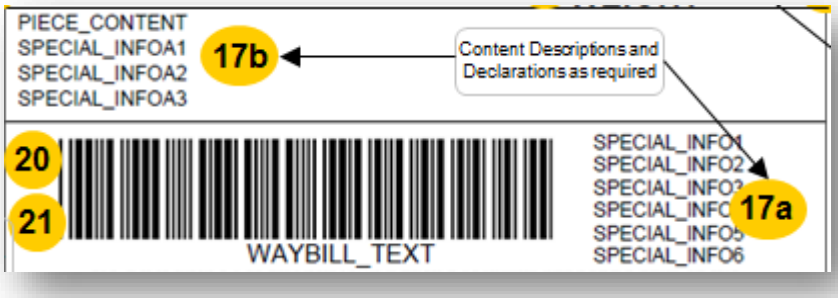
5.13. Additional Shipment Information

Customers may wish, in certain cases and for various reasons, to add and/or print additional information (e.g. a country-mandatory Export Declaration) on the label. The same may apply for internal reasons.

The Global Label Standard is equipped with distinct segments that cater for the need of additional information.

[9] Refer to sections 5.18 and 8.1 for further details on this capability and related specifications.

The image below identifies those info elements (which extend the format, previously limited to the area adjacent – segment 17a - to the Waybill Barcode).



1.1.1. Content Description (mandatory)

The **set of segments (17)** is reserved for such information and consists of:

- (17a) placed right to the Waybill barcode – supports 150 chars
- (17b) which goes across the whole label width – supports 300 chars

The font size is to be chosen, within the physical constraints specific to each template version, in such a way that at least 300 chars can be printed in both sections 17a and 17b.

Note: practical usage so far has taught that keeping **segment 17a associated with the content description** is a good approach, in case there are additional requirements for regulatory information (see example below, where the regulatory information was placed in segment 17b).

The **following practical guidance** is to be observed:

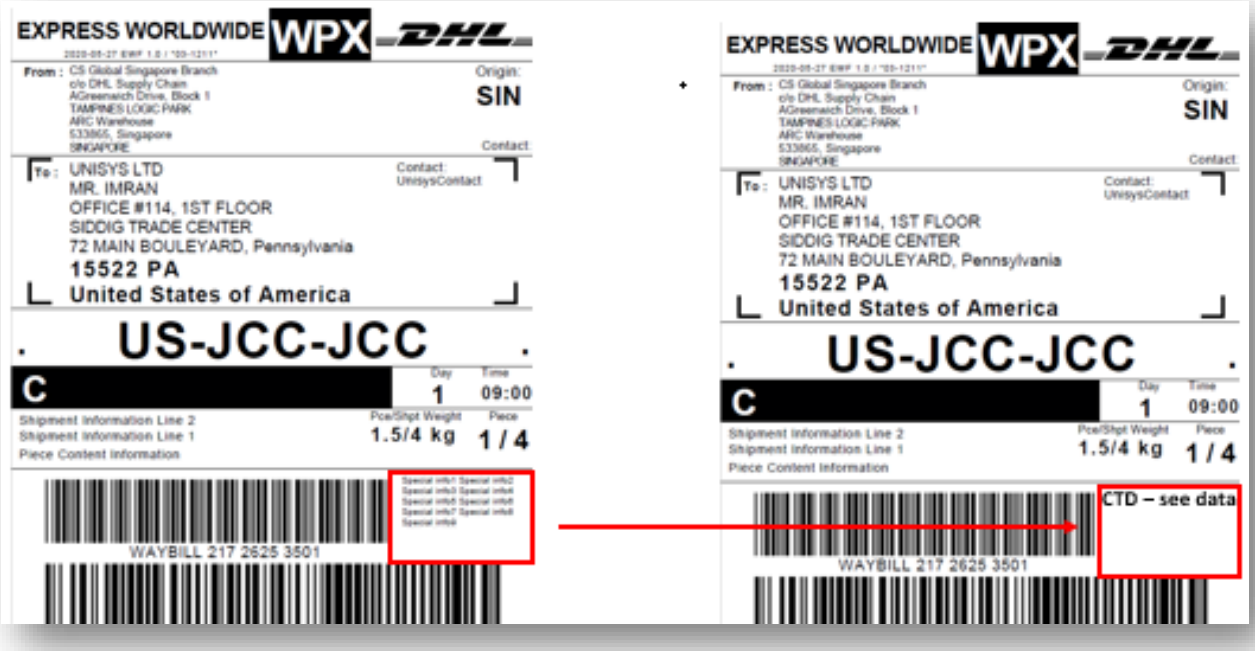
- **"Content Description"** or related Information (**mandatory**)
  - **To be located primarily in segment 17a**, starting in Line 1 ("Special_Info1)
  - **Exceptional scenario**: Should there be a scenario where the required content description extends beyond the pre-defined number of characters for segment 17a, then IF there is no Regulatory Information using segment 17b, the content description can be located in the segment 17b
- **General Regulatory Information**
  - To be located **in Segment 17b**, starting in Line 2 (Special_InfoA1)

1.1.1.1. Neutral Content Descriptions

There is a requirement from specific shippers, who wish to “neutralize” the Goods Description on the Transport Label to help with what they perceive as a security threat during physical shipment handling.

For selected and approved customers, who meet the defined Neutral Content Label criteria, the respective content descriptions on the Transport Labels can be replaced with a “Neutral Content Description” text, defined as “**CTD – see data**”). The “**CTD – see data**” string must be printed with bold font and with a min 3mm character height.

The location of the “Neutral Content Description” complies with the guidance of the previous section (“Content Description”).



1.1.2. Special or Conditional Information

- **Special/ Conditional Information (see examples below)**
  - To be, as a guiding principle, located in segment 17b
  - **Dangerous Goods Declarations are to be located in segment 17a**
  - The guiding principle follows however the primacy of the General Regulatory Information, in which case the Special Information, if existent, can be moved to the “Piece Content” line (refer to first image, above, in this section). It would then also take precedence over any piece-related contents.
  - Should two data elements need to be printed in the “Piece Content” line (e.g. they can be concatenated in the same line if separated by specific captions and a “/” separator (e.g. CNP-12345678 / IE-987654321)
    - A Tax ID can also be understood and represented as a Shipment Reference with the associated Reference Type, in which case it would not be placed here but rather in the Piece / Shipment Reference segment.

1.1.3. Not allowed on Transport Labels

- **Account Numbers / Customs Value / Shipment Insurance Value <- must be excluded**
  - Neither Account Numbers nor Customs Value or Shipment Insurance Values are allowed on the Transport Label
  - Moreover, any data elements not explicitly stated and required on Global SOP Standard Transport Label are, by definition, not allowed and must be excluded.

5.13.1. Conditional Local Information

In some countries, additional information is legally required on the label, usually for a small percentage of the shipping volume. Any information of this kind may be added to these specifications.

Depending on the size of the information required, it may be inserted (sometimes making the label longer or even requiring a separate label to be used). Any potential separate label has to carry the Piece Identifier in human-readable form and respective barcode in addition. Wherever such a case fits into one of the existing segments it is described in the respective context.
All other cases are described in this chapter.

5.13.1.1. Tax IDs (Brazil) - Conditional

Labels for packages shipped from addresses in Brazil need to display the Tax IDs (Called CNPJ/CPF and IE/RG) for both origin and destination: this is an approved exception, strictly subject to central endorsement, through the DHL EXPRESS Label Certification Body.

Upon DHL EXPRESS' approval, these numbers need to be printed as part of the respective addresses (if free address lines are available) or to be printed in segment 17, with at least the font size of the consignee address.

Note: In this special case (BR), the TAX ID can also be understood and represented as a Shipment Reference with the associated Reference Type, in which case it would not be placed in this segment but rather in the Piece / Shipment Reference segment.

5.13.2. Dangerous Goods and Other Special Loads

Dangerous goods and other special loads (Live Animals – Perishable Cargo – Human Remains) by law require additional information that may be printed in the shipment information segment or as a separate label.

DHL Express does not offer an unconditional transport of dangerous goods and other special loads. Prohibited Commodities are excluded from transportation.

The "Prohibited and Restricted Commodities Policy[10]" defines, categorizes and lists what can and cannot be transported by DHL EXPRESS. Goods are therein defined into two categories:

- Prohibited
- Restricted

[10] The definition and listing of "Prohibited and Restricted Commodities" are subject, at any time, to updates through the corresponding Regulators. Please always refer to the latest International Standard version for use purposes.

For additional information on **Mandatory** / **Conditional** Legal Entries required on the eWaybill or Transport Label for Special Loads, refer to the following documents (DHL Guidelines and Regulations):

- *Prohibited and Restricted Commodities Policy -> DP-DHL Policy*
- *ICAO/IATA DG – IATA AVI – IATA PER – IATA HUM Regulations: International regulations for the transport of Dangerous Goods, Live Animals, Perishable Cargo and Human Remains by Air*
- *ADR: European Agreement concerning the International Carriage of Dangerous Goods by Road*
- *IMDG code: International Maritime Dangerous Goods Code for the transport of Dangerous Goods by Sea*
- *RID : Regulations concerning the International carriage of Dangerous Goods by Rail*

5.14. Shipment Identifier / Waybill Number

5.14.1. Shipment Identifier / Waybill Number (mandatory)

The Shipment Identifier (proxy for the Waybill Number) is mandatory for all DHL Express products on the Transport Label. Its omission requires explicit approval by DHL.

The Shipment Identifier must be printed in groups of four digits, with the grouping sequence starting at the end and being built towards the left side (equivalent to the mandatory grouping of the Piece Identifier).

5.14.2. Shipment Identifier / Waybill Number barcode (Conditional)

The Waybill Number barcode is always printed in the Standard Code 39 symbology with the syntax and human-readable format as specified in the DHL EXPRESS Global SOP Barcode Specifications.

See section 8.2.2 for technical Code 39 specifications and chapter 7 for DHL-specific requirements.

5.15. Routing Information (mandatory)

The DHL ISO Routing Code contains destination country, destination Postcode, transport products and associated product features. When a routing field is **optional**, the content is not required.

*In the context of the routing barcode, **optional** does not always imply that the routing field may remain empty. Some routing fields must always be populated at least with zeroes. If data is available, though, such a field should always be filled.*

The Postcode, the region specific fields and the + separator after the product/feature block are the only **optional** fields which must be omitted if the data is not available.

5.15.1. Routing Barcode Structure

The following international ISO standards are supported when coding routing information in barcodes.

- The **ASC MH 10** routing barcode starts with the ASC MH 10 Data Identifier "**2L**" and is coded in barcode symbology Code 128.
- The **GS1** routing barcode starts with the GS1 Application Identifier "**403**" and is coded in barcode symbology GS1-128.

Both standards are complementary and can co-exist without restriction (they do not differ in content but only in their technical structure). ***However, DHL Express-internal applications will only produce ASC MH 10 routing barcodes starting with 2L.***

A distinction has to be made between the information which will be captured and the information, which will actually be printed out on the barcode. This distinction is mainly relevant for applications which need an indication on the output side.

Example: At data content level, a calendar date for delivery is not chosen for the selected product, so the delivery date information will not be captured. The delivery date is optional, as it does not always have to be captured. However, at barcode level, the delivery date digits are **mandatory** as the delivery date field may not remain empty/blank and the field must be populated with zeroes if no valid information is available.

*All routing information appears both in human and bar-coded format except for the **parenthesis around the Data Identifier** which only appear in human readable format – not in the barcode.*

The printout level is either **mandatory** or **optional**, it means the whole barcode is either printed or not printed (with the possible exception of a country-specific extension).

A product/feature block always consists of 8 digits, regardless of which products and features have been chosen.

The table below details out ASC MH and GS1 barcode routing structures

SECTION	ASC MH 10	GS1	String
LEFT PARENTHESIS	(		Mandatory ***(only in plain text – not in barcode!)***
DATA / APPLICATION IDENTIFIER	**"2L"**	**"403"**	Mandatory
RIGHT PARENTHESIS	)		Mandatory ***(only in plain text – not in barcode!)***
ISO COUNTRY CODE	**2 upper-case alpha chars**	**3 numeric digits**	Mandatory For domestic shipments as well
POST CODE	Postcode of Consignee address. **Variable length, max. 12 chars**; The Postcode may never contain spaces or symbols **Remains empty in case of country with no Postcode system.** In case of known DHL Service Area Code (3 chars) or a DHL Delivery Facility Identifier (6 chars), this can be coded here, separated from the country code by a colon ":".		**Conditional** (Mandatory where existent)
FIELD SEPARATOR	"+" char		Mandatory
PRODUCT CODE	**2 numeric digits (fixed length)**		Mandatory
DELIVERY DATE	**2 numeric digits (fixed length)**		Mandatory If no date specified, put two zeroes.
DELIVERY TIME	**1 numeric digit (fixed length)**		Mandatory (if no delivery time, put one zero)
HANDLING FEATURE CODE(S)	**3 digit numeric number with leading zeroes (fixed length)** Sum of values of product features		Mandatory (if no feature, put three zeroes)
FIELD SEPARATOR	**FNC1** (Special Code 128 symbol, in all three code sets.)	**"+"** char	**Optional** May remain empty
COUNTRY-SPECIFIC CODES	Variable up to 6 numeric chars or up to 2 alphabetic chars, **for domestic use only**.		**Optional** May remain empty

Example of a routing code: 2LBE3500+11311002123456

(2L)	BE	3500	+	11	31	1	002	123456
Data Identifier	ISO Country Code	Postcode code	Field separator	Product code	Delivery Date	Delivery Time	Product Feature	Region specific codes

When none of the above features is chosen, the code is “000”.

5.15.2. Full Routing Barcode Examples

a) ASC MH 10
EUROPACK piece, without any features but with region specific routing information:

	Country	Postcode code		Product	Date	Time	Feature	Domestic routing code
2L	**DE**	**81541**	**+**	**05**	**00**	**0**	**000**	**556677**
	Germany			DOM EUROPACK	No fixed date	No fixed time	No features	Street and house number

ECONOMY SELECT shipment, to be held at depot, dutiable:

	Country	Postcode code		Product	Date	Time	Feature
2L	**CH**	**3000**	**+**	**57**	**52**	**0**	**001**
	Switzerland			ECONOMY SELECT	Hold at depot	No fixed time	Customs Clearance

EXPRESS WORLDWIDE shipment, to be delivered on the 10th (future feature), in time code 1, with customs clearance (code 001) and two future product feature with codes 016 and 064:

	Country	Postcode code		Product	Date	Time	Feature
2L	**NL**	**1023FG**	**+**	**48**	**10**	**1**	**081**
	The Netherlands			EXPRESS WORLDWIDE	Delivery at 10th of month	Time code 1	81= 1+16+64

ECONOMY SELECT shipment, to be held at depot, dutiable:

	AI	Country	Postcode		Product	Date	Time	Feature
FNC1	**403**	**756**	**3000**	**+**	**57**	**52**	**0**	**001**
Start char		Switzerland			ECONOMY SELECT	Hold at depot	No fixed time	Customs Clearance

5.16. Customer Segment

This section is reserved and can be used by the shipper (or customer) for information and own purposes (e.g. for parties involved in the logistics chain). Its standard location is directly above the Piece Identifier segment. These contents are transparent for DHL.

DHL imposes no restrictions in size or content of this segment, as long as the information printed in it does not interfere with any other information on the label.

This section may contain barcodes (see sample image below) but none are allowed that could be interpreted as one of the following content types:

- DHL Waybill Number (10 digits, numeric, barcode in Code 39)
- DHL Routing Code (Data Identifier 2L / Application Identifier 403)
- Piece Identifier (Data Identifier J, 1J …, nJ / Application Identifier 00)

NOTES. The objective of the highest-possible level of standardization is valid also for this segment. In specific exceptional business scenarios, upon customer requirement, a relocation of the Customer Segment is possible, requires however an explicit approval from DHL EXPRESS.

- The potential new location must remain in the lower area of the label, below the Handling Feature Codes' segment, close to the existing barcode segments, ideally directly above segment 20 (Waybill barcode)
- A different approach is to locate the customer-owned segment beyond the boundaries of the standard layout. In this case, the customer-owned segment is no longer part of the label and shall not be defined herein. To reinforce that fact, DHL requires that a thick line, with a min of 2 mm, be drawn to visually separate the specific boundaries of the label from external elements. To prevent detrimental effects on the PID scanning performance, only locations above the label upper boundary are acceptable.

5.17. Piece Identifier Segment

Each piece of each shipment shipped with DHL requires a Piece Identifier as its unique identifier. Any ISO 15459-compliant Piece Identifier may be used.

5.17.1. Content

The Piece Identifier:

- must not contain more than 35 chars (excluding the Data Identifier “J”)
- may only contain numeric and upper case alphabetic chars drawn from ISO/IEC 646 – they must not include any lower case char or punctuation mark
- must start with a string of chars representing an issuing agency assigned by ISO

DHL EXPRESS issued Piece Identifiers (LPs) usually begin with either a “JD00…“ or a “JD01…” format prefix.

In the specific case of the JD01 LP-types, its length i.e. the number of characters following the Data Identifier “(J)” is exactly 20 – see example, below.

5.17.2. Barcode and Plain Text

A Piece Identifier must be printed as a barcode. Its content must be repeated in plain text below the barcode.

Although, according to the official standard, both Code 128 and Code 39 are allowed to encode Piece Identifier numbers, any barcode produced by a CAS system or any DHL internal software has to use Code 128.

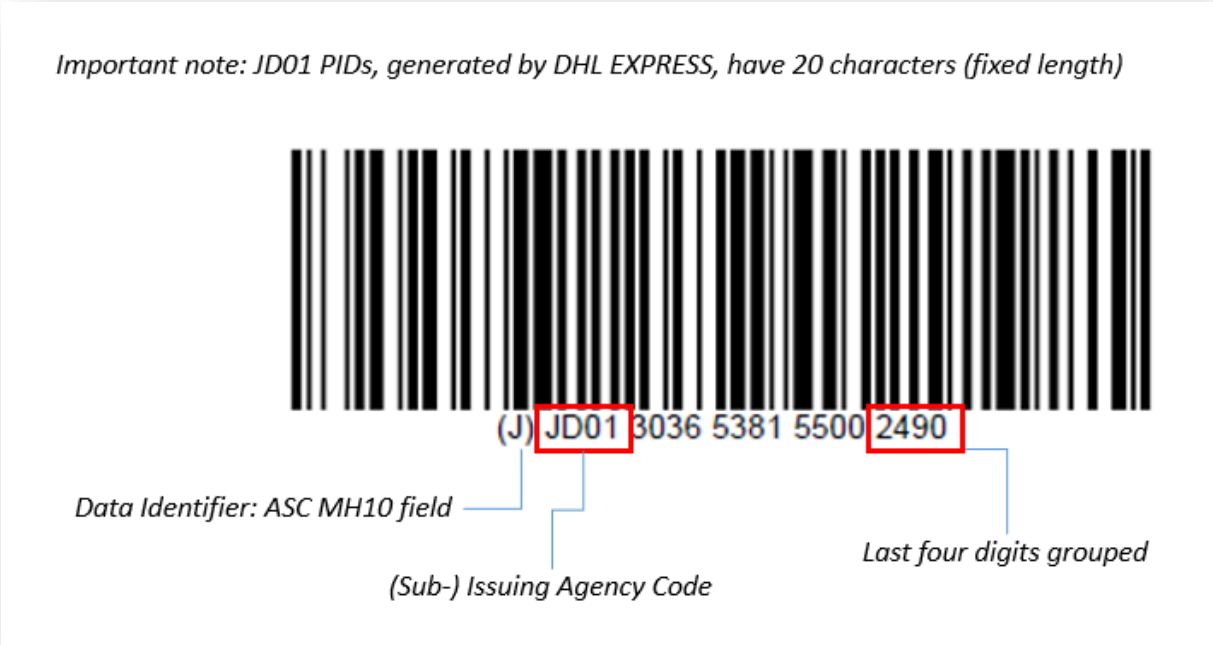
5.17.3. Data/Application Identifier

The Data Identifier (for ASC MH 10 data fields) or Application Identifier (for GS1 data fields) is an ISO-specified string of chars that identifies the content of the data that follows.

All DHL issued Piece Identifiers will be of type ASC MH10, using Data Identifier “J”.

For readability reasons of ASC MH10 Piece Identifiers, DHL EXPRESS requires that the human-readable code be grouped in such a way that blocks of four digits follow the (Sub-) Issuing Agency Code. Those are built upwards (from left to right) so that the last four digits also form a group.

*This way of printing Piece Identifier in groups of four digits is **mandatory** for all DHL-owned customer automation software.*



The Data Identifier preceding the Piece Identifier is NOT part of the Piece Identifier. It does not have to be considered when determining the Piece Identifier length. Two Piece Identifiers are identical if their only difference is a deviating data identifier (e.g. "J" versus "2J").

However, the Data Identifier always has to be printed on the label both in human readable and bar-coded format. The bar-coded Data Identifier must not include the parentheses which are used for clarity only.

(The bar-coded form of the data identifier and Piece Identifier is then "JJD01……". & the human readable form should be in the form "(J)JD01 …..".)

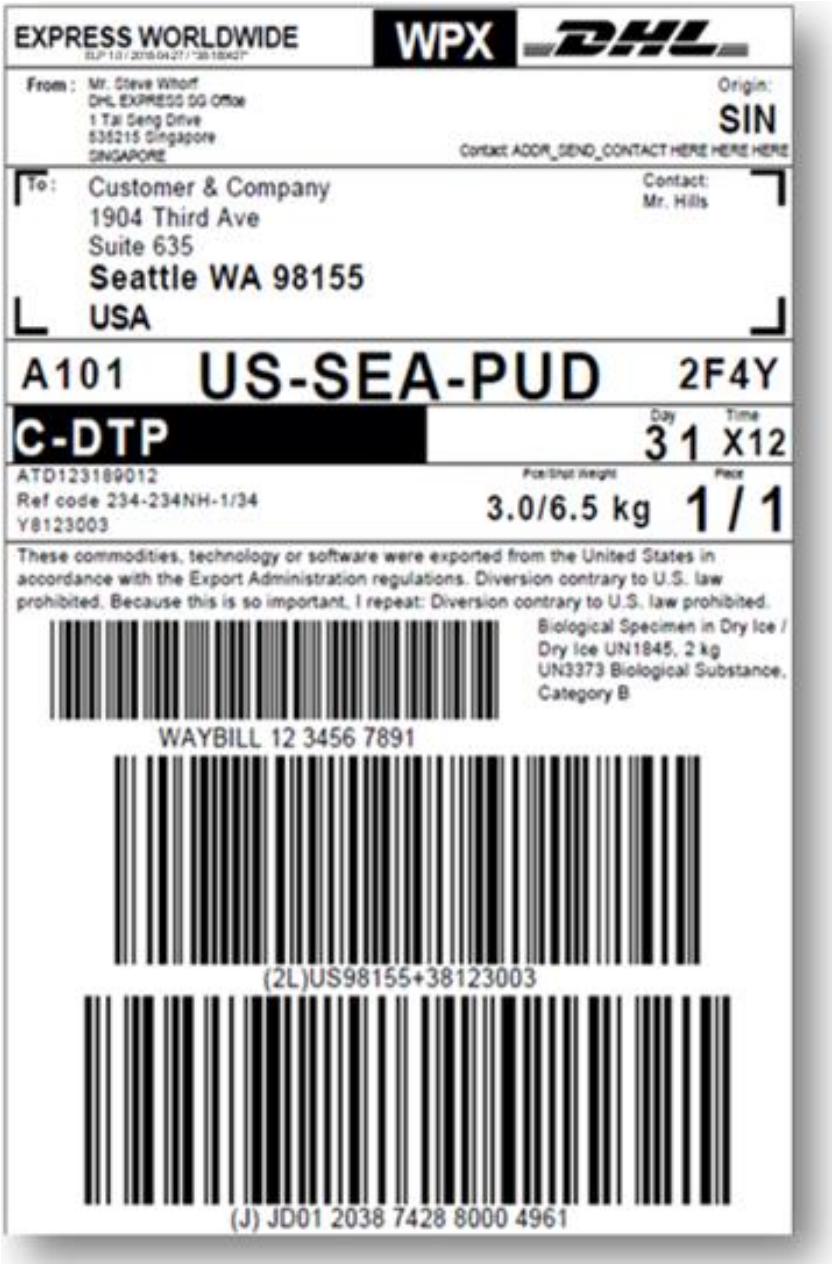
5.18. Sample Labels

The following labels serve solely as an illustration of the label structure and content.

5.18.1. ESS Label V2.8 - Size 6"x4" / A6

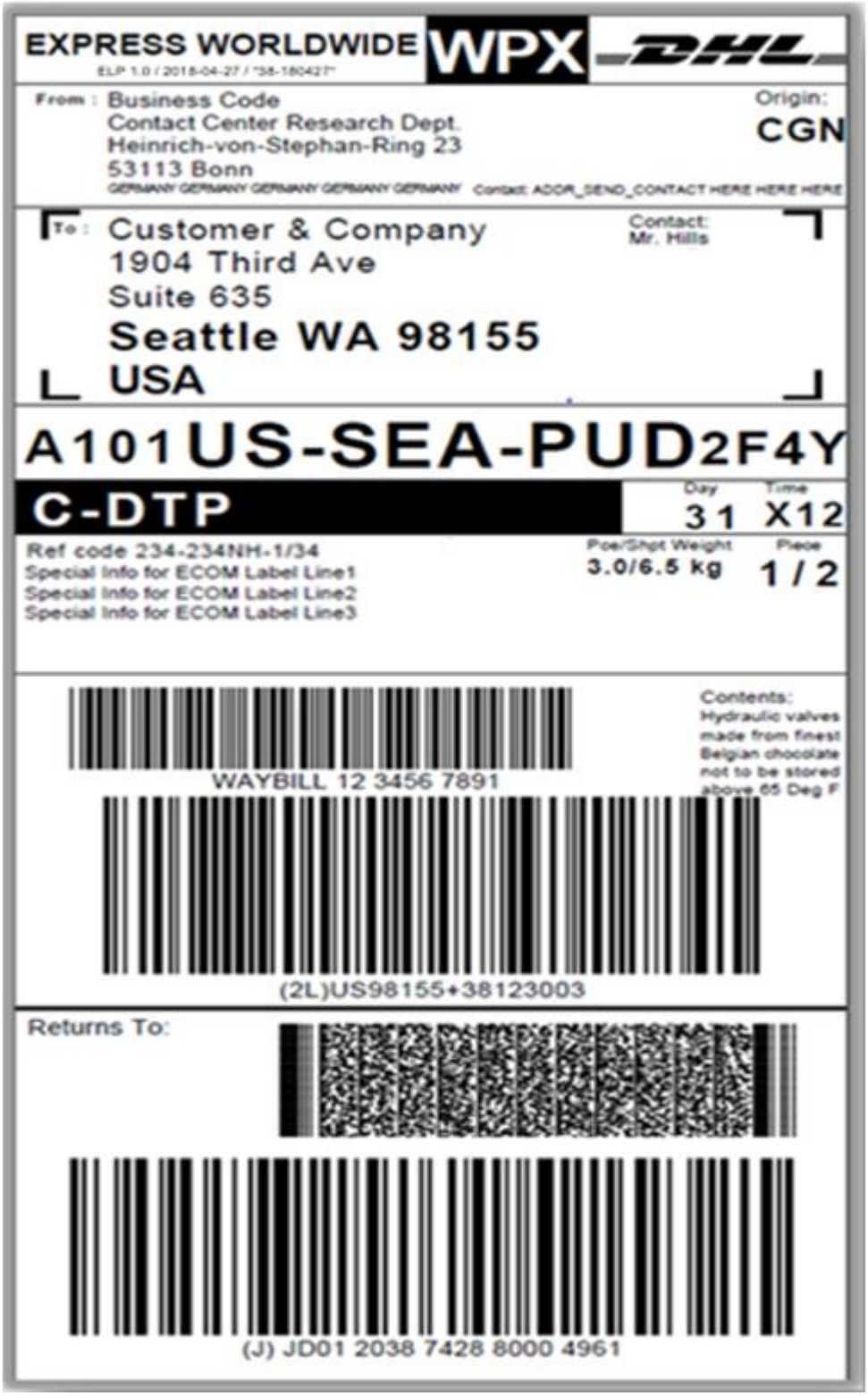
Example with multiple Services

This DHL A6 ESS Label (length: 160 mm / width: 100mm) should be the format (length) chosen to guarantee best-possible balance of compactness and quality for scanners.

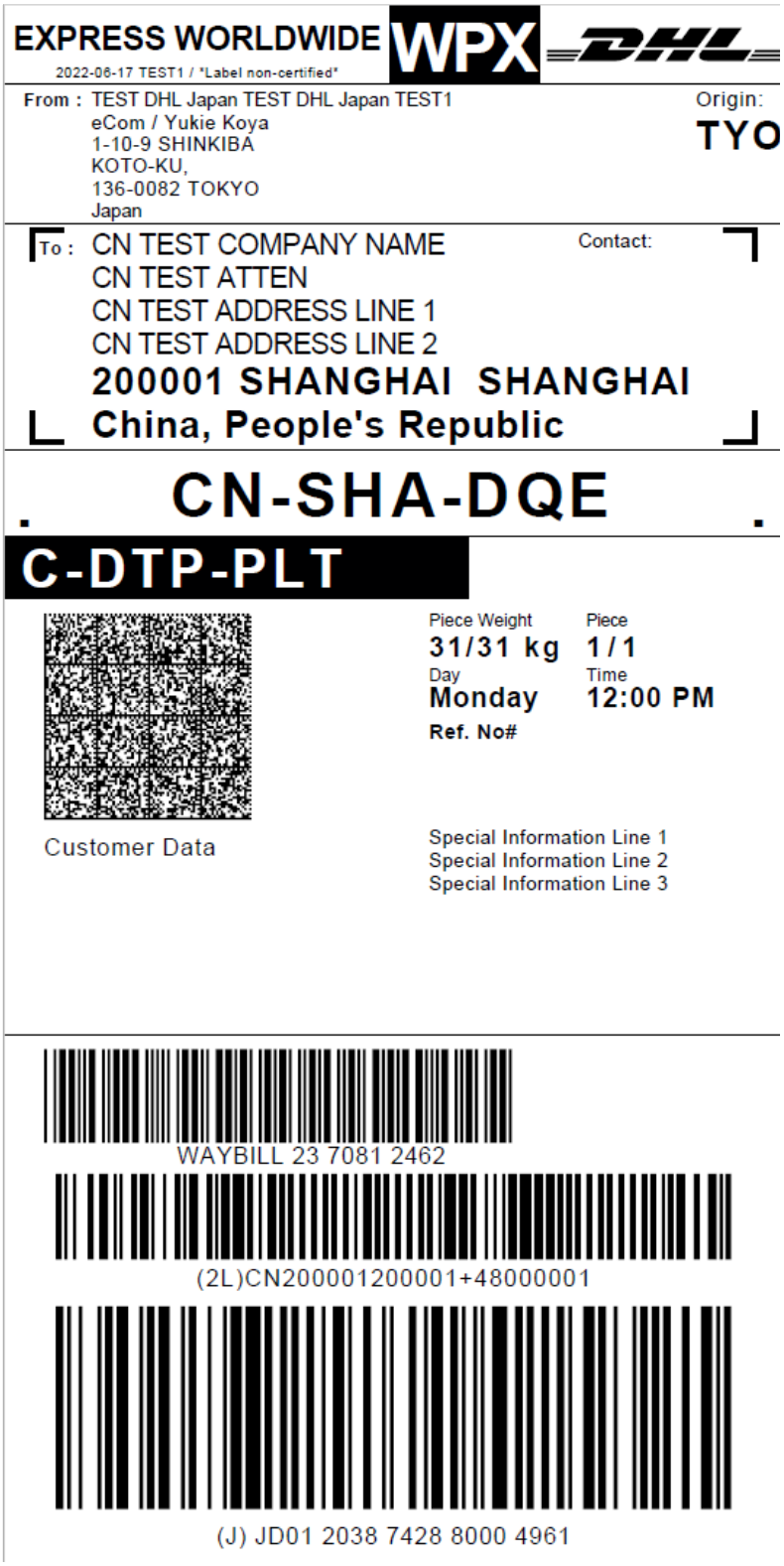


5.18.2. ESS Label V2.8 – Size 8”x 4” w. Customer Info

Where the label length is not subject to other constraints, a length of 8 inches easily allows for an additional segment with customer-specific information. If required, the label can even be made longer than this.


EXPRESS WORLDWIDE
WPX
From : Business Code
Contact Center Research Dept.
Heinrich-von-Stephan-Ring 23
53113 Bonn
Origin:
CGN
To : Customer & Company
1904 Third Ave
Suite 635
Seattle WA 98155
USA
Contact:
Mr. Hills
A101 US-SEA-PUD 2F4Y
C-DTP
Day
31
Time
X12
Ref code 234-234NH-1/34
Special Info for ECOM Label Line1
Special Info for ECOM Label Line2
Special Info for ECOM Label Line3
Pce/Shpt Weight
3.0/6.5 kg
Piece
1 / 2
Contents:
Hydraulic valves made from finest Belgian chocolate not to be stored above 65 Deg F
WAYBILL 12 3456 7891
(2L)US98155+38123003
Returns To:
(J) JD01 2038 7428 8000 4961

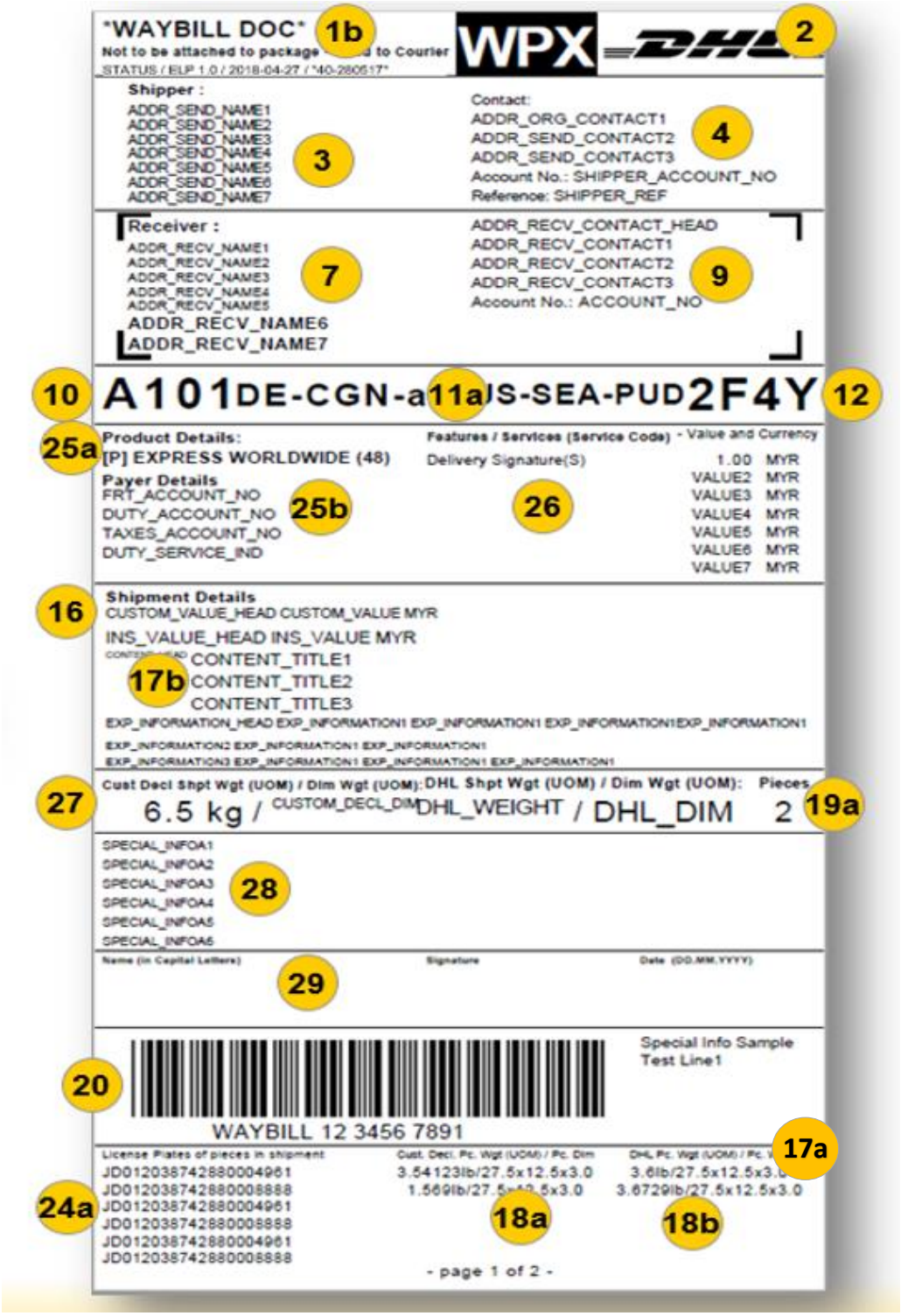
5.18.3. ESS Label V3.0 – Size 8”x 4” w. 2D symbol (Customer Data)



6. eWaybill – Segment Specifications

A **Global Standard eWaybill** is required for all shipments, either in printed or electronical form, whenever the associated paperwork needs to be archived (imaged) and indexed. It is a Document type containing all information and data relevant for correctly billing a customer or supporting Data Entry, Billing-related Invoice Validation processing or even to be used as a Customer Receipt.
Due to distinct functions taken by the eWaybill Type and audiences, the Reference Purpose is to be printed as working Title (see image below) and shall vary accordingly (e.g. “Waybill”, “DHL Copy – please hand to Courier”, “Customer Receipt”).
This is a document type at shipment level, which is never to be appended (or visible in) to a package because it is not a transport label! It is also not required for all products in printed form – see Global SOP for further details.

6.1. eWaybill – Layout Overview (all segments[11])



[11] This page offers a structured overview of the Global SOP Standard Electronic Waybill (eWB). Sub-sequent sections explain in detail the specifications of each individually numbered segment.

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Note: Only the segments which are not common to both the Global Standard Label and the “eWaybill”, will be defined and described in the following sections.

The eWaybill in this specification has been designed to allow for both Transport Label and eWaybill printing on the same printer. It is, however, also allowed to print the eWaybill on a normal A4 / Letter size document, provided that all specified information is available and the Waybill Barcode is printed.

6.2. Purpose – Working Title of Document (optional)

The **intended purpose** (e.g. “eWaybill Doc”) of the document will be printed here, with an alpha-numeric format.

This information indicates the purpose of the document, which may vary depending on the intended use and/or targeted user type (e.g. Customer or DHL Staff). More than one purpose can be selected at the time of rendering, with the result that “several” documents with distinct “purposes” are created.

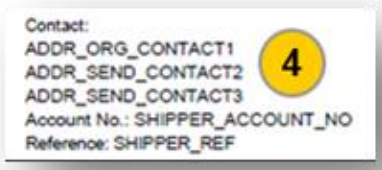
Examples of the targeted users and associated “purposes” are:

- **DHL OPS -> “DHL Copy – please hand to Courier”**
- **Customer -> “Customer Copy”**
- **DHL Billing -> “Waybill Doc”**

The format of this segment is:

- **[“Waybill Doc”]**
- **[INTENDED PURPOSE]**
- **[STATUS]”/”[PRINTING APPLICATION]”/”[PRINTING DATE]”/” [CERTIFICATION IDENTIFIER]**
  - where “Status” can have the following values and will be controlled by a specific parameter:
    - **“Validated by DHL”**
    - **“” -> left blank**
  - Printing Application
  - Printing Date: the document print date must be printed according to the Standard ISO 8601 (YYYY-MM-DD).
  - where the “Certification Identifier” adopts the same syntax of the Transport Label (see 5.2.3)

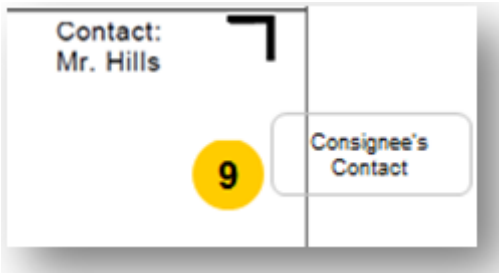
6.3. Sender’s Contact



The Sender's Contact information in this segment supports the following optional elements in addition to the label (refer to chapter 4):

- Local language translation of Contact Name
- Account Number – Identifies the Shipper's Account Number
- Reference – Identifies the set of Piece/ Shipment References

6.4. Consignee's Contact



The Consignee's Contact information in this segment supports the following optional elements, in addition to the label (refer to chapter 4):

- Local language translation of Contact Name
- Account Number – Identifies the Consignee's Account Number

6.5. Country-Facility ID

The DHL Country Code and Facility Identifier for both Original and Destination locations will be displayed here, with the following format:

- [ORIGIN COUNTRY CODE]-[ORIGIN SERVICE AREA CODE] [ORIGIN FACILITY CODE], in correspondence to the DHL Facility managing the Shipper's GEO Area

- [DESTINATION COUNTRY CODE]-[DESTINATION SERVICE AREA CODE] [DESTINATION FACILITY CODE], in correspondence to the definition applied to the corresponding Transport Label and to the DHL Facility managing the Consignee's GEO Area

6.6. Product Information

6.6.1. Product Details (mandatory)

The Product chosen by the customer will be printed here, with the following data elements and format:

- [Network Product Code] Global Product Name (Product Sort Code)

Referring to the example above, the product is being identified with the following elements:

- "[P]" – NETWORK PRODUCT CODE
- "EXPRESS WORLDWIDE" – GLOBAL PRODUCT NAME
- "(48)" – PRODUCT SORT CODE

6.6.2. Payer Details (mandatory)

Payment-related information, captured by the shipping applications in association with the specific shipment scenarios corresponding to the payment codes, will be presented with the following format:

, where:

- "FRT:" (FREIGHT) PAYER ACCOUNT NUMBER (depending on service)
- "DHL Billing Service:" BILLING SERVICE INDICATOR
- "Duty Payer:" DUTY PAYER ACCOUNT NUMBER
- "Taxes Payer:" TAX PAYER ACCOUNT NUMBER
    - where both Identifiers always match* and depend on the service selected
    - *DHL does not split between Duty and VAT Billing and related accounts, so they are always equivalent. For this reason, a single line "Duty/ VAT A/C" is also acceptable

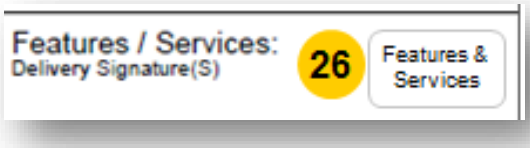
The Service selected in combination with the account number in the Duty Payer A/C# and Tax Payer A/C # fields, determines who will pay what.

DTU – with no account number mentioned is the default DHL service and Consignee always pays.

Where the Consignee account number is required but not yet known (e.g. in case of the Basic Feature "Duty and Taxes Unpaid"), the line will just state "Duty: REC / Tax: REC".

6.7. Features and Services

DHL will consider, by default, that all context-related Charge Codes are applicable except for the so-called Basic Features (i.e. the ones being part of the product).



All services as ordered by the customer, to be grouped into one or more columns (see example next) to accommodate the following data elements:

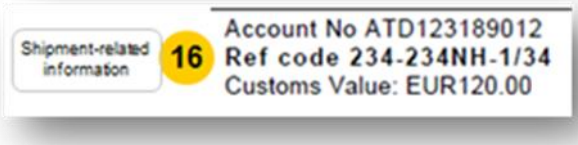
"GLOBAL SERVICE NAME"	"PRODUCT SERVICE CODE"	"VALUE OF SERVICE"	"CURRENCY CODE"
Delivery Signature	S	1 000.00	EUR

Some systems use specific/ distinct formats for values and decimal digit representations. For this reason, the format "# ### ### ###.###" will be used hereafter as standard for currency values.

Charge Amount values will follow the format 1999.01 as "1[space]999.01", where a single "space" will have the role of a digit group separator in place of a comma.

The Currency Code shall comply with the ISO 4217 Currency Code Standard.

6.8. Shipment-related Information

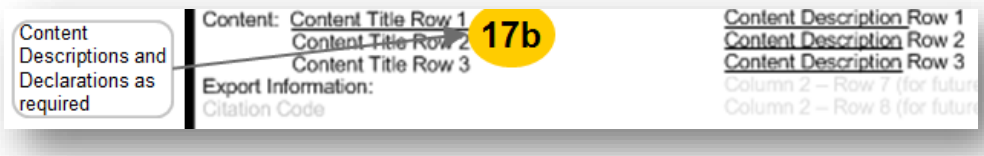


Additional Shipment-related information may be added to this segment as required, in which case additional lines, within the pre-defined limit, can be used.

The following data elements are foreseen [13](and possible) as per DHL EXPRESS Standards:

- "Declared Value for Customs" – CUSTOMS VALUE
  - indicates the declared customs value, if non-doc shipment
  - Format: <# ### ### ###.##>, where currency code is **mandatory**
- "Declared Value for Insurance:" – DECLARED VALUE FOR INSURANCE
  - Format: <# ### ### ###.##>, where currency code is **mandatory**
  - Default value is that of the Shipper's Country Currency Code
- "Commodity ID:" - COMMODITY IDENTIFIER
  - as per DHL's Global Commodity Code Standard
- "Commercial Invoice Number:" – COMMERCIAL INVOICE NUMBER
  - as per DHL's Global Commercial Standards

6.9. Content Descriptions and Declarations



This segments 17a and 17b contain information on the shipment contents, their description and regulatory declarations. Additional data elements or information may be added.

[13] Contrary to the Global Standard Transport Label, where non-piece-handling data elements are to be omitted (by default), here, all data elements supporting the full description of the Customer's Transport Order are wished for. The above structure has been closely aligned with Global Billing and other key stakeholder teams.

The format is the following:

- "Content 1:" – <CONTENT TITLE ROW 1><CONTENT DESCRIPTION ROW 1>
- "Content 2:" – <CONTENT TITLE ROW 2><CONTENT DESCRIPTION ROW 2>
- …
- "Export Information:" – EXPORT INFORMATION
- "EEI:" – ELECTRONIC EXPORT INFORMATION (OR INDICATOR)

Regulatory declaration in regards to Dangerous Good information is mandatory and is to be located in **Segment 17a**, starting in Line 1 (SPECIAL_INFO1).

6.10. Weight – Declared and DHL-Measured

6.10.1. Shipment Weight (mandatory)

The "shipment weight", legacy term corresponding to the addition of the individual piece weights, is to be printed in this segment, with the following format:

- [SHIPPER DECLARED SHIPMENT WEIGHT (UOM)] / [DIMENSIONAL WEIGHT (UOM)]

- [DHL SHIPMENT WEIGHT (UOM)] / [DIMENSIONAL WEIGHT (UOM)]
  - Content optional -> subject to availability of value at the time of rendering

Its value must correspond to the dead weight plus package dimensions OR the applicable volumetric weight, to allow for charge calculation based on that document.

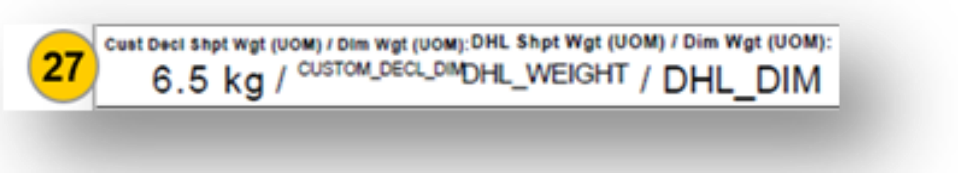
Should the volumetric weight NOT be available at the time of rendering the document, then it may be omitted.

Weight, as provided by the shipper (e.g. "Shipper Declared Weight" through EDI / 2D Code / Manual Waybill), has to be printed on the label IF the printing occurs prior to DHL Reweigh processing, in this field.

Should, however, labeling happen after weighing, then DHL label printing applications must print the determined weight ("DHL Actual Weight") instead. CAS systems won't be able to print the "DHL Actual Weight".

When no weight was available at the time of printing the label, a "." shall be printed to indicate that fact.

The mass unit must be printed next to the weight.

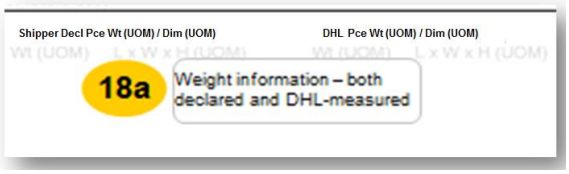


6.10.2. Piece Weight

The piece weight of the individual pieces is to be printed in this segment with the following format:

- [SHIPPER DECLARED PIECE WEIGHT (UOM)] / [DIMENSIONAL WEIGHT (UOM)]
- [DHL PIECE WEIGHT (UOM)] / [DIMENSIONAL WEIGHT (UOM)]
  - Content optional -> subject to availability of value at the time of rendering

Its value must correspond to the dead weight plus package dimensions OR the applicable volumetric weight, to allow for charge calculation based on it.



The mass unit must be printed next to the weight.

6.11. Piece Counter

The total number of pieces belonging to the shipment shall be printed here.

This information is conditional to its reliability. Shall its correctness not be guaranteed, then it is to be suppressed. In this last scenario:

- the customer must mandatorily provide the information by other reliable means (e.g. manifesting message)
- upon DHL approval, the information may be temporarily suppressed on account level.
  - an approved Global SOP Exemption Request shall document the agreed data provision process and the expected improvement approach

6.12. Special Information

Special information is to be printed here; it corresponds to the “Special Info” conveyed by the Transport Label. There is a 1:1 association between both document types to this regard. The same number of individual lines (9) will be supported in both documents to ensure backward compatibility.

Special Information in this segment refers primarily to content-related facts (e.g. “Belgian chocolate“ or “Li-metal batteries with equipment”).

Should the content-related information require longer line(s), it may be combined with segment 17b (“Contents and Descriptions”).

6.13. Customer's Signature (optional)

In the event that this document type is used, as a Proof of Collection (Receipt) for B2C Returns to DHL EXPRESS, **when required by the Customer**, the Courier's signature is to be captured here, together with the Date and Time. The signed eWaybill copy is then handed to the Shipper and the other copy is taken back for imaging in the station.

6.14. Piece Identifiers in Shipment (mandatory)

The exhaustive list of pieces belonging to the shipment shall be identified here.
The Piece Identifiers shall be printed here in plain text format.

7. LOOSE BBX LABEL SPECIFICATIONS

7.1. Purpose

The purpose of this document is to supplement the current Global SOP procedures with finer detail relative to the specification, design, production and use of a Network-compliant Loose BBX Transport Label.

7.2. Objective

This guideline shall ensure that stakeholders:

- understand What a Loose BBX Transport Label is and its purpose, in the context of DHL Express business segment
- know How to create a Loose BBX Transport Label and How to use it
- understand and comply with the business requirements for Label data and information
- understand the flexibility allowed by the design and its corresponding constraints
- are able to identify Label issues or non-compliance and initiate mitigation actions

7.3. Scope

This reference guide is directed at all DHL Functions and Facilities that are involved in the production or use of Loose BBX Transport Labels and corresponding transportation and handling of Shipments, with specific reference to the:

- DHL Loose BBX Transport Label
- All related Architectural elements (e.g. Identifiers, Barcodes)
- Label Certification Guidelines

7.4. Layout Overview (all segments)

The circled coloured numbers in the image below correspond to individual segments, which are described in detail in the following sections.

NOTE: the Loose BBX Transport Label corresponds, in its essence, to a standard Transport Label, where a new Custom's Routing Address segment, identified through the new sub-segments 25-27, was added. Thus, the below layout will only cover the added sub-segments, everything else remains standard (refer to page 12 for related definitions).

Product Name and Content Name
1
BREAKBULK
BBX
2
DHL Logo
From : THAI PRODUCTS
1700/114 Ocean Road
Khlong Tamru, Muang
3
0200 Bangkok
THAILAND
5
LOGO
text
6
Origin:
BKK
DHL's Service Area Code of Origin
4
Sender's Contact
Import Port Of Entry
25
IMPORT AT :
THAI PRODUCTS
c/o DHL LEJ HUB
Hermann-Koehl Strasse
04435 SCHKEUDITZ
GERMANY
Contact:
26
Import PoE:
DE-LEJ-LEJ
DHL Gateway Facility ID
Total Nr. pieces: 385
27
MHWB: 0987654321
HWB: 12345678901
pieces, Mother HWB, (Baby) HWB
To : YELLOW CIRCLE
Ben Johnson
Av. Alvaro Duarte
41092 SEVILLA
SPAIN
Consignee's Address
7
8
LOGO
text
Contact:
9
Consignee's Contact
DHL Destination Country-Facility ID
11
Outbound Sort Code
10
A10 ES-XVQ-XVQ 2F4Y
12
Inbound Sort Code
Handling Feature Code
13
C-DTP-LBB
14
Day
31
Time
X12
15
Delivery Time (of day)
Shipment-related information
16
Ref code: 234-234NH-1/34
18
Pce/Shpt Weight
3.0/6.5 kg
Piece
1 / 2
19
Piece Counter (per Shipment)
SPECIAL_INFOA1 / UN1845 Dry Ice, Net 5 kg, UN3373 Biological substances - Category B
Content descriptions and Declarations, as required
17b
Calendar Day, for delivery
22
DHL's Routing Code
(2L)US98155+38123003
License Plate (Piece Identifier)
24
(J) JD01 2038 7428 8000 4961

7.5. Import Point of Entry - Mandatory

This segment contains the address of DHL's Import Clearance Gateway Address with the caption "IMPORT AT:" This address segment is standard, with the same specifications as the other two (Shipper and Receiver).

7.6. DHL Gateway Facility Identifier

This segment contains DHL's Gateway Facility Identifier, with the caption "Import PoE", acronym for "Import Point Of Entry". This element is compliant with DHL's Reference Data for Facility identification.

7.7. Mother House Waybill Number (MHWB) - Mandatory

This segment contains the Master House Waybill Number in human-readable format, with two related data elements, in total:

- Total Number of Pieces for the Mother, with corresponding caption "Total Nr. Pieces:" – optional field
- Mother HWB in human-readable form, with corresponding caption "MHWB" – mandatory field
- (Baby) House Waybill number in human-readable form, with caption "HWB" - mandatory field

8. Specifications of Label Types, Fonts and Sizes

This paragraph describes all defined types of DHL Automated Labels, each of them with mandatory and optional sections including size and position on the label.

The values for font sizes specify minimum font sizes. However, the maximum height per block has to be considered. DHL's document rendering tool (GLS) introduced a dynamic management feature for the font size, to ensure information does not get unnecessarily truncated in case of long contents, which may lead to slightly distinct font sizes within the different lines of a common segment. This is to be seen as functionally correct even if at the expense of graphical representation consistency.

Furthermore, the **font size for the Consignee address** must be bigger than the font size for the Shipper address. For each address, the Postcode and City/Town font size must not be smaller than the size of the biggest font used in the respective address lines.

For all font sizes, the values given for "Font height in mm" and "Size in dots at 200 dpi" are rounded values, provided solely for ease of programming. The size always includes the space between the lines.

Although **ARIAL** font type is recommended, other fonts can be accepted if they are without serifs and if they provide equal legibility.

The default and preferred orientation for the DHL label is the portrait orientation. A landscape format is only possible as an exemption, to be approved by DHL's Central Label Certification Office. DHL-owned applications are not allowed to print landscape format labels.

Font sizes are subject to two conflicting requirements: max font size for readability and min font size for compactness. For this reason, in scenarios where a high compactness degree is required (e.g. customer with much data in the customer-owned segment) the font size for the Shipper Address lines can be decreased to a min font size of 1.5 [mm]. Accordingly, the font size for the Receiver Address lines may be decreased to a min font size of 2.5[mm]. In both special cases, a formal approval from the Global Label Certification Center is required (mandatory).

In the segment descriptions, thin red lines are used to separate segments. These lines must not be printed. A thick red line shows where a black line needs to be printed as a segment separator on the label.

8.1. DHL ESS Label v2.8

8.1.1. Size and Orientation

The label width must be **within 95 and 110 mm**. This requirement is met by both A6 labels (portrait) and 4" wide labels.

The preferred length of a DHL Transport Label is between **148 mm (A6 length) and 152.4 mm (6")**, but it can be longer if necessary to support requirements for additional information. **A general maximum length is not defined**. However, each customer must be given the possibility to print labels that are not longer than 250 mm. Depending on the piece dimensions, no label may be too long to fit on one side of the package.

As an exception, the label can be divided in two smaller sized labels. In this case, the label that does not contain the Piece Identifier barcode form must show the unique piece identifier at least in human readable form (minimal font size 8 point).

CAS *If a printout on A5 or Statement format (i.e. folded A4 or folded Letter) is desired, the label content can be divided into two columns, with the first column to contain the addresses plus potential*

> *customer-specific information and the second column to be identical to a Network Label (see **Error! Reference source not found.**)*

8.1.2. Segment Fonts and Sizes

Product Name 1 Line 1: char height 3mm, max 20 char, bold Line 2: char height 1 mm	**Content Code** Char height 6mm (17pt)	**DHL Logo** 2 Min width 30mm. Keep original ratio May be replaced with "DHL" for non-graphics printer	
Shipper Address 3 7 lines max, min 4. Char height (6pt) Postcode, City Name, State/Province (bold font) Country (bold font), max 20 char	**Customer Logo** 5 Max width 30 mm. Must not overlap Shipper Address	6 **Area Code of Origin /** 4 **Contact Info** Origin: char height 2mm (6pt) DHL Service Area Code: 3 char, 3mm hight, bold	
Consignee Address 7 7 lines max, including Country Name and Contact Details; all w. min 3 mm char height (9pt) Country, Postcode, City Name, State/Province 4mm char height (bold font), max 35 char Country (bold font), max 20 char	**Consignee's Logo** 8 Max width 20 mm. Must not overlap Consignee's Address	**Contact Details** 9 All lines 2mm, char height (6pt)	
Outbound Sort 10 Left-adusted, max 5 mm from edge Char height: 5-6 mm (14pt)	**Destination Facility Code** 11 Centered, char height: 7-9 mm recommended 9 mm (20 pt)	**Inbound Sort** 12 Right-adusted, max 5 mm from edge Char height: 5-6 mm (14pt)	
Services / Handling Features (HFC) 13 Bold white on black background (inverse video), cahr height 6mm (17 pt). HFC string has variable length. Reduce font to size 8, if string does not fit into field i.e. 15 chars (including separators)		**Delivery Date** 14 Header: 2 mm char height (6 pt) Value: 4 mm char height (11 pt)	**Delivery Time** 15 Header: 2 mm char height (6 pt) Value: 4 mm char height (11 pt)
Shipment Information 16 Line 1: Shipment reference (optional) - 2 mm char height (6 pt) Lines 2-3: 2 mm char height (6 pt)	**Weight** 18 Header: 2mm char height (6 pt) Weight in Kg/ lb, 4 mm char height (11 pt), bold font		**Piece Count** 19 Header: 2 mm char height (6 pt) Value: 4 mm char height (11 pt)
Special Information (if required) 17b All lines w. 2mm char height (6pt) Contents (optional): Line 1 - Piece Info; lines 2-4 - Special Info			
20 **Waybill (Shiment Identifier) Barcode / 2D Code** 21 Symbology: Code 39 - see barcode specs section for further details Symbol height: 11 mm, with min 5 mm quite zone on both left/ right sides Text below barcode: "WAYBILL" in capital leters, followed by Waybill No. /Shipment ID, 3 mm char height 2D Code - refer to 2D Code specs		**Contents / Special Info (e.g. DG, if required)** 17a All lines 2mm, char height (6pt)	
Routing Code 22 Symbology Code 128 for ASC MH 10 Application Identifier "2L"; GS1-128 for Application Identifier "403" - - see barcode specs section for further details Symbol height: 11 mm, with min 5 mm quite zone on both left/ right sides The full barcode length must be under 91 mm. Text below barcode: full contents w. DI/AI in parenthesis, w. 3mm char height (9pt)			
Customer-owned Information 23 No restrictions other that overall label dimensions. Information (also barcoded) shall not interfere with DHL contents. It may include text, graphics, images.			
License Plate (Piece Identifier) Barcode 24 Symbology Code 128 for ASC MH10 License Plates; GS1-128 for SSCC License Plates - see barcode specs section for further details Min quite zones: 5mm (left and right sides). Barcode height must be equal or higher than 25mm, length must be under 91 mm Min x-dim must not be less than 0.33 mm and not higher than 0.51 mm. Text below barcode: full contents w. DI/AI in parenthesis, w. 3mm char height (9pt)			

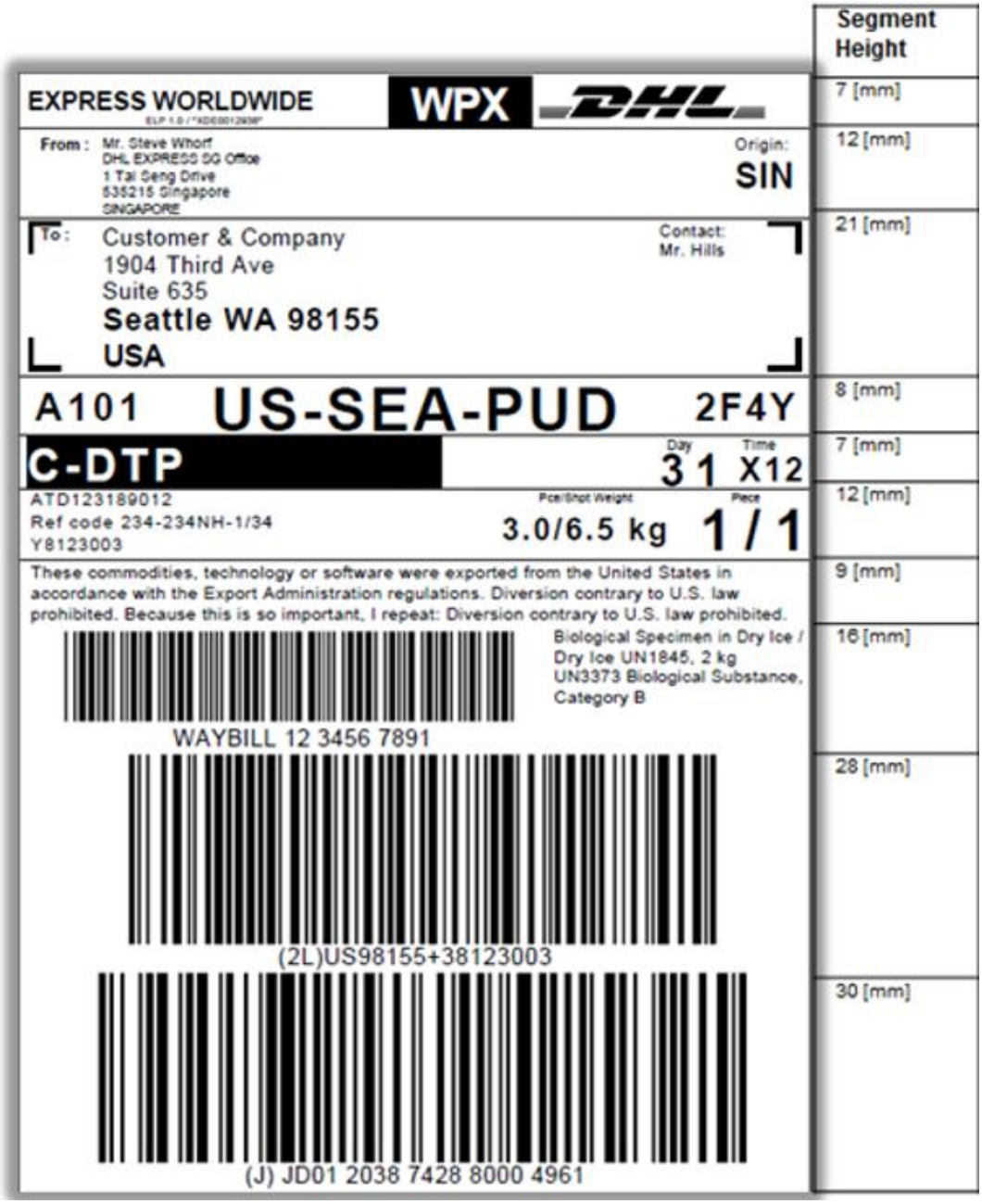
8.1.3. Considerations on A6 or 6”x4” (inch) Labels

The ESS Label v2.8 fits on an A6 label (14.85 cm) and on a 6”x4” (15.24 cm) label.

However, the design has to be done very carefully, and the printing will not allow for huge tolerances. Users in the USA are generally recommended to use 6 ¾” x 4” labels.

The following table is a good indicator for a best-possible layout of an A6 label.

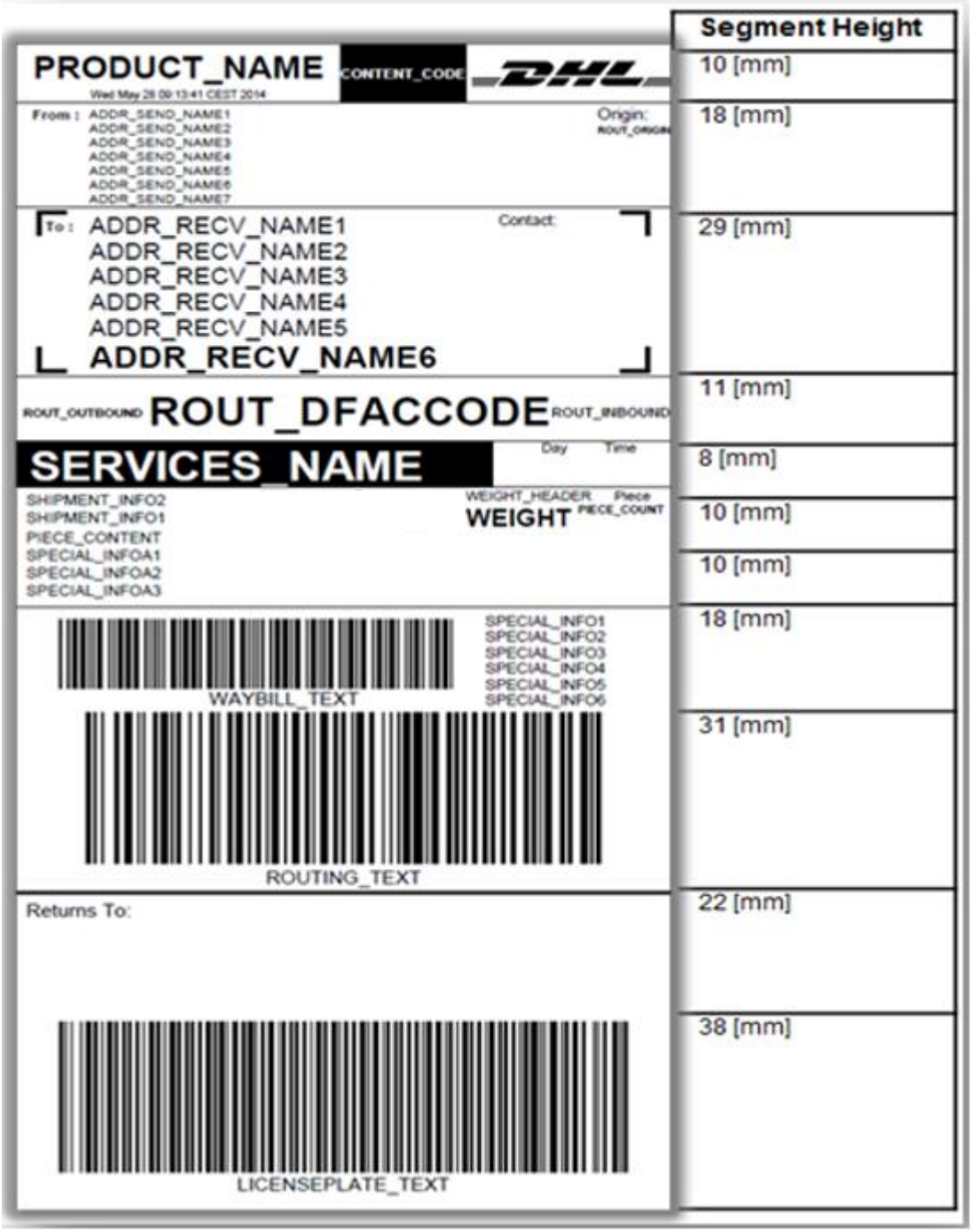
Note: this challenge with space exists only when the Waybill Barcode is present on the label.



This works slightly easier with 6” long labels which are about 4 mm longer than A6 labels.

8.1.4. Considerations on labels for 8” or A4 landscape print

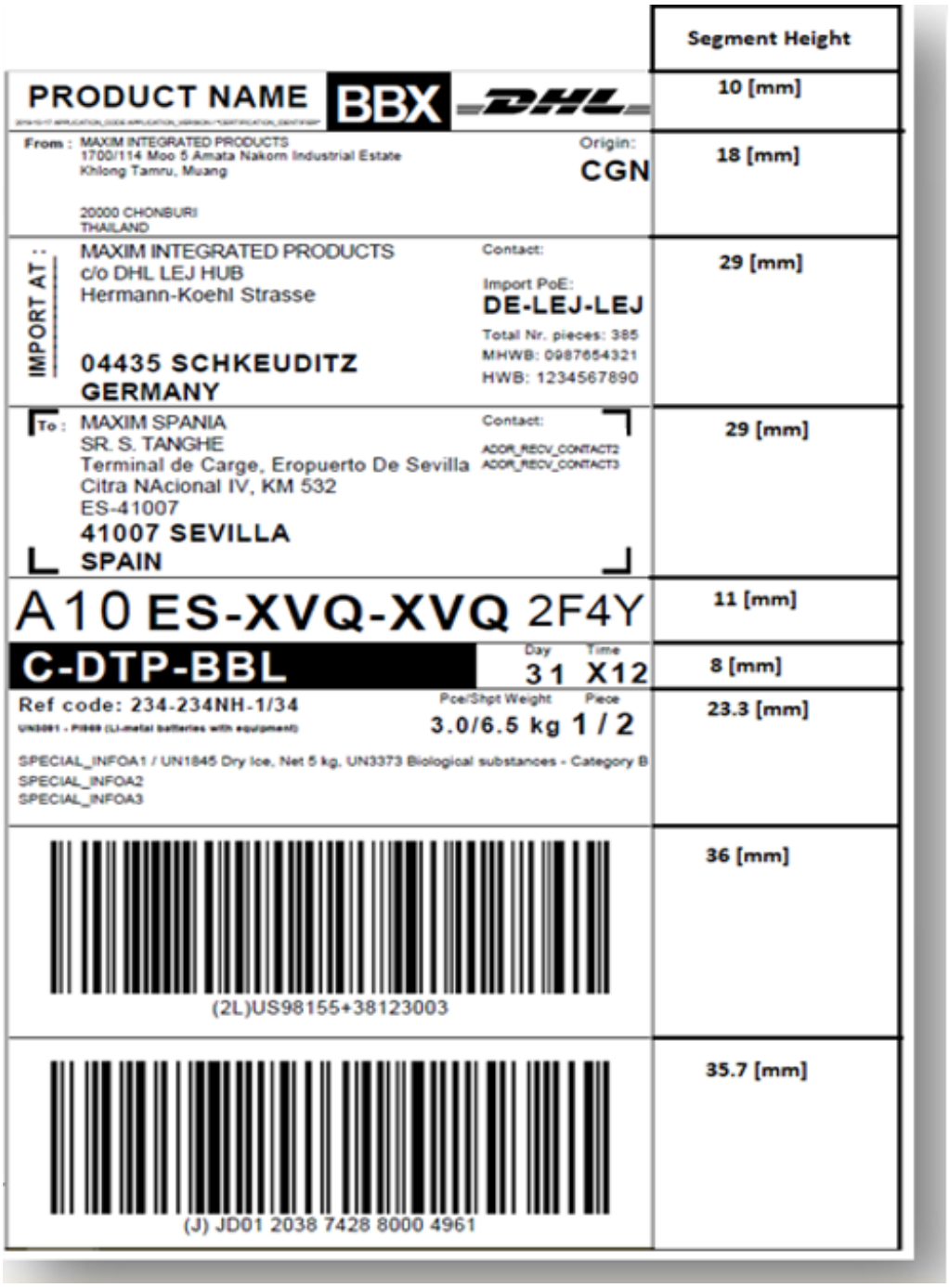
This section gives the recommended segment and font sizes for v2.8 labels as they should be used on thermo printers where possible. The length of 8” is equivalent to 20.3 cm which makes this a good fit for A4 or letter format sheets in landscape format. One sheet can contain two or (in the case of A4) three labels, or one label plus further information.



8.1.5. Considerations on Loose BBX Transport Labels

This section gives the recommended segment and font sizes for “Loose BBX Transport Labels” as they should be used on thermo printers where possible.

These Labels may exist with the standard 8” format. Its usage is subject to a formal Global Label Certification.



8.1.5.1. Segment Fonts and Sizes

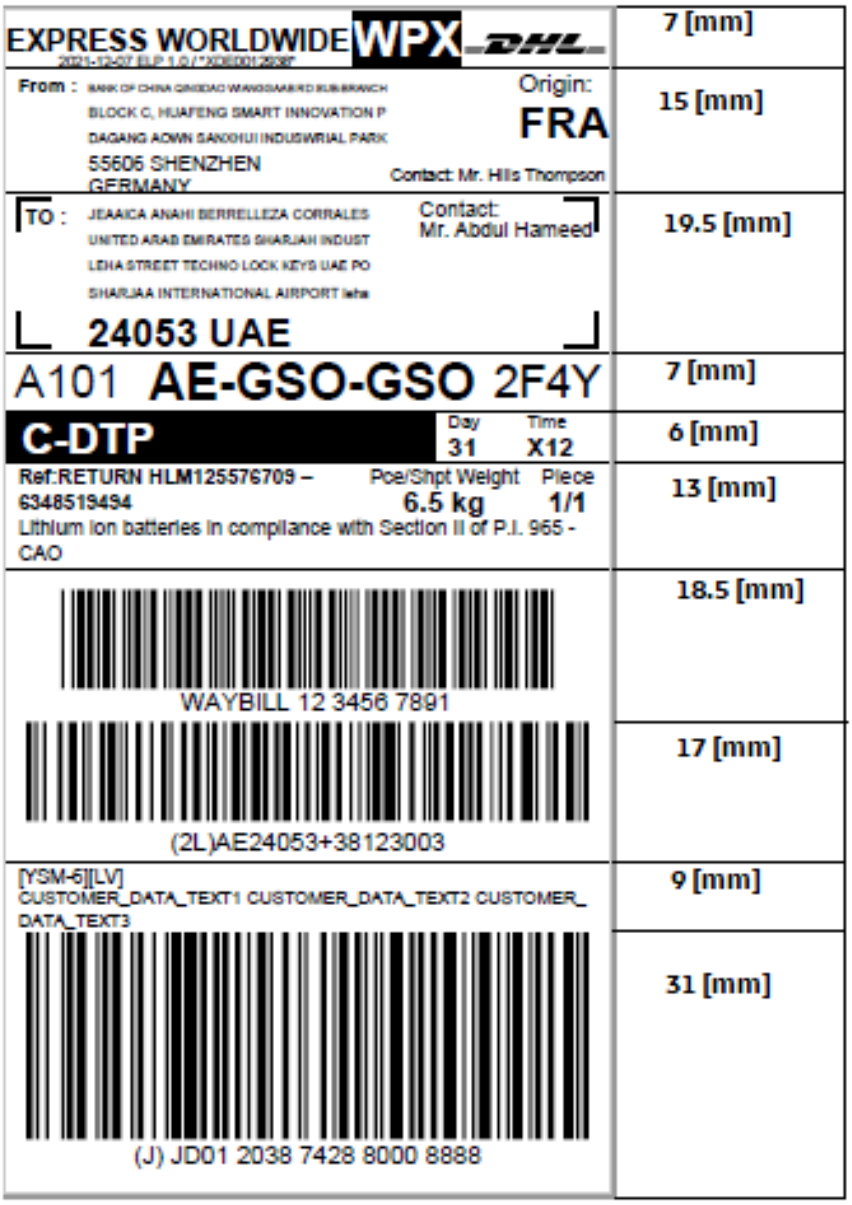
With the exception of the new sub-segments 25-27, all fonts and corresponding sizes remain standard.

Product Name 1 Line 1: char height 3mm, max 20 char, bold Line 2: char height 1 mm	**Content Code** Char height 6mm (17pt)	**DHL Logo** 2 Min width 30mm. Keep original ratio May be replaced with "DHL" for non-graphics printer	
Shipper Address 3 7 lines max, min 4. Char height (6pt) Postcode, City Name, State/Province (bold font) Country (bold font), max 20 char	**Customer Logo** 5 Max width 30 mm. Must not overlap Shipper Address	6 **Area Code of Origin /** 4 **Contact Info** Origin: char height 2mm (6pt) DHL Service Area Code: 3 char, 3mm height, bold	
IMPORT AT (caption) - vertically oriented (90 degrees, counterclockwise), char height 3 mm, **Importer of Record's Address** 7 lines max, including Country Name and Contact Details; all w. min 3 mm char height (9pt) Co 25 Postcode, City Name, State/Province 4mm char height (bold font), max 35 cha Country (bold font), 4mm char height, max 20 char		**DHL Gateway Facility Identifier** **Contact Details** All lines 2mm, char height (6pt) 26 **DHL Gateway Facility ID**, with caption "**Import PoE**", 2 mm char height **DHL Gateway Facility ID**, 8 standard char, with 3 mm char height "**Total Nr. Pieces** 27 **MHVB**" and "**HVB**" captions,	
Consignee Address 7 7 lines max, including Country Name and Contact Details; all w. min 3 mm char height (9pt) Country, Postcode, City Name, State/Province 4mm char height (bold font), max 35 char Country (bold font), max 20 char	**Consignee's Logo** 8 Max width 20 mm. Must not overlap Consignee's Address	**Contact Details** All lines 2mm, char height (6pt) 9	
Outbound Sort 10 Left-adusted, max 5 mm from edge Char height: 5-6 mm (14pt)	**Destination Facility Cod** 11 Centered, char height: 7-9 mm recommended 9 mm (20 pt)	**Inbound Sort** 12 Right-adusted, max 5 mm from edge Char height: 5-6 mm (14pt)	
Services / Handling Features (HFC 13 Bold white on black background (inverse video), cahr height 6mm (17 pt). HFC string has variable length. Reduce font to size 8, if string does not fit into field i.e. 15 chars (including		**Delivery Date** 14 Header: 2 mm char height (6 pt)	**Delivery Time** 15 Header: 2 mm char height (6 pt)
Shipment Information 16 Line 1: Shipment reference (optional) - 2 mm char height (6 pt)	**Weight** 18 Header: 2mm char height (6 pt) Weight in Kg/ lb, 4 mm char height (11 pt), bold font		**Piece Count** 19 Header: 2 mm char height (6 pt)
Special Information (if required) 17b All lines w. 2mm char height (6pt) Contents (mandatory): Line 1 - piece contents; lines 2-4 - Special Info (optional)			
Routing Code 22 Symbology Code 128 for ASC MH 10 Application Identifier "2L"; GS1-128 for Application Identifier "403" - - see barcode specs section for further details Symbol height: 11 mm, with min 5 mm quite zone on both left/ right sides The full barcode length must be under 91 mm.			
Customer-owned Informatio 23 No restrictions other that overall label dimensions. Information (also barcoded) shall not interfere with DHL contents. It may include text, graphics,			
License Plate (Piece Identifier) Barco 24 Symbology Code 128 for ASC MH10 License Plates; GS1-128 for SSCC License Plates - see barcode specs section for further details Min quite zones: 5mm (left and right sides). Barcode height must be equal or higher than 25mm, length must be under 91 mm Min x-dim must not be less than 0.33 mm and not higher than 0.51 mm. Text below barcode: full contents w. DI/AI in parenthesis, w. 3mm char height (9pt)			

8.1.6. Considerations on Highly-Compact (HC) Labels

This section gives the recommended segment and font sizes for “Highly-Compact - HC” labels as they should be used on thermo printers where possible.

HC-Labels’ usage is subject to a formal Global Label Certification.



8.1.6.1. Segment Fonts and Sizes

Product Name 1 Line 1: char height 3mm, max 20 char, bold Line 2: char height 1 mm	**Content Code** Char height 5mm (17pt)	**DHL Logo** 2 Min width 20mm. Keep original ratio May be replaced with "DHL" for non-graphics printer	
Shipper Address 3 5 lines max, min 3. Char height (6pt) Postcode, City Name, State/Province (bold font) Country (bold font), max 20 char		6 **Area Code of Origin /** 4 **Contact Info** Origin: char height 2mm (6pt) DHL Service Area Code: 3 char, 3mm hight, bold	
Consignee Address 7 5 lines max, including Country Name and Contact Details; all w. min 3 mm char height (9pt) Country, Postcode, City Name, State/Province 4mm char height (bold font), max 35 char Country (bold font), max 20 char		**Contact Info** Contents line: 2mm, char height 9	
Outbound Sort 10 Left-adusted, max 5 mm from edge Char height: 5-6 mm (14pt)	**Destination Facility Code** 11 Centered, char height: 7-9 mm recommended 9 mm (20 pt)	**Inbound Sort** 12 Right-adusted, max 5 mm from edge Char height: 5-6 mm (14pt)	
Services / Handling Features (HFC) 13 Bold white on black background (inverse video), char height 5mm (17 pt). HFC string has variable length. Reduce font to size 8, if string does not fit into field i.e. 15 chars (including separators)		**Delivery Date** 14 Header: 2 mm char height (6 pt) Value: 3 mm char height (11 pt)	**Delivery Time** 15 Header: 2 mm char height (6 pt) Value: 3 mm char height (11 pt)
Shipment Information 16 Line 1: Shipment reference (optional) - 2 mm char height (6 pt) Lines 2-3: 2 mm char height (6 pt)	**Weight** 18 Header: 2mm char height (6 pt) Weight in Kg/ lb, 4 mm char height (11 pt), bold font		**Piece Count** 19 Header: 2 mm char height (6 pt) Value: 4 mm char height (11 pt)
Special Information (if required) 17b All lines w. 2mm char height (6pt) Contents (optional): Line 1 - Piece Info; line 2 - Special Info			
20 **Waybill (Shiment Identifier) Barcode** Symbology: Code 39 - see barcode specs section for further details Symbol height: 11 mm, with min 5 mm quite zone on both left/ right sides Text below barcode: "WAYBILL" in capital leters, followed by Waybill No. /Shipment ID, 3 mm char height 2D Code - refer to 2D Code specs			
Routing Code 22 Symbology Code 128 for ASC MH 10 Application Identifier "2L"; GS1-128 for Application Identifier "403" - - see barcode specs section for further details Symbol height: 11 mm, with min 5 mm quite zone on both left/ right sides The full barcode length must be under 91 mm. Text below barcode: full contents w. DI/AI in parenthesis, w. 3mm char height (9pt)			
Customer-owned Information 23 No restrictions other that overall label dimensions. Information (also barcoded) shall not interfere with DHL contents. It may include text, graphics, images.			
License Plate (Piece Identifier) Barcode 24 Symbology Code 128 for ASC MH10 License Plates; GS1-128 for SSCC License Plates - see barcode specs section for further details Min quite zones: 5mm (left and right sides). Barcode height must be equal or higher than 25mm, length must be under 91 mm Min x-dim must not be less than 0.33 mm and not higher than 0.51 mm. Text below barcode: full contents w. DI/AI in parenthesis, w. 3mm char height (9pt)			

8.1.7. Considerations on 2D-enabled Transport Labels

This section gives the recommended segment and font sizes for “2D-enabled Transport Labels” labels as they should be used on thermo printers where possible and other printer types when allowed.

2D-enabled Labels may exist with the usual standard formats (8” or 6”) or new ones (e.g. customer-tailored, upon DHL EXPRESS approval and subject to a formal Global Label Certification).

8.1.7.1. Segment Fonts and Sizes

Product Name 1 Line 1: char height 3mm, max 20 char, bold Line 2: char height 1mm	**Content Code** Char height 6mm (17pt)	**DHL Logo** 2 Min width 30mm. Keep original ratio May be replaced with "DHL" for non-graphics printer
Shipper Address 3 7 lines max, min 4. Char height (6pt) Postcode, City Name, State/Province (bold font) Country (bold font), max 20 char	**Customer Logo** 5 Max width 30 mm. Must not overlap Shipper Address	6 **Area Code of Origin /** 4 **Contact Info** Origin: char height 2mm (6pt) DHL Service Area Code: 3 char, 3mm hight, bold
Consignee Address 7 7 lines max, including Country Name and Contact Details; all w. min 3 mm char height (9pt) Country, Postcode, City Name, State/Province 4mm char height (bold font), max 35 char Country (bold font), max 20 char	**Consignee's Logo** 8 Max width 20 mm. Must not overlap Consignee's Address	**Contact Details** All lines 2mm, char height (6pt) 9
Outbound Sort 10 Left-adusted, max 5 mm from edge Char height: 5-6 mm (14pt)	**Destination Facility Code** 11 Centered, char height: 7-9 mm recommended 9 mm (20 pt)	**Inbound Sort** 12 Right-adusted, max 5 mm from edge Char height: 5-6 mm (14pt)
Services / Handling Features (HFC) 13 Bold white on black background (inverse video), cahr height 6mm (17 pt). HFC string has variable length. Reduce font to size 8, if string does not fit into field i.e. 15 chars (including		

Data Matrix 2D Barcode 25 Symbology "Data Matrix ECC 200"; see barcode specs section for further details Min quite zones: not less than 2 modules, but 5mm recommended (all sides) when feasible. Customer Data symbol: Min x-dim must not be less than 0.38 mm and not higher than 0.40 mm. Text below barcode: 3mm char height (9pt) Auto-Sorting symbol: Min x-dim must not be less than 0.76 mm and not higher than 1.0 mm. Text below barcode: 3mm char height (9pt)	**Weight** 18 Header: 2mm char height (6 pt) Weight in Kg/ lb, 4 mm char height (11 pt), bold font	**Piece Count** 19 Header: 2 mm char height (6 pt)
	Delivery Date 14 Header: 2 mm char height (6 pt) Value: 4 mm char height (11 pt)	**Delivery Time** 15 Header: 2 mm char height (6 pt)
	Shipment Information 16 Line 1: Shipment reference (optional) - 2 mm char height (6 pt) Lines 2-3: 2 mm char height (6 pt)	
	Contents / Special Info (e.g. DG, if required) 17a All lines 2mm, char height (6pt)	
	Special Information (if required) 17b All lines w. 2mm char height (6pt) Contents (optional): Line 1 - Piece Info; lines 2-4 - Special Info	

20 **Waybill (Shiment Identifier) Barcode / 2D Code** Symbology: Code 39 - see barcode specs section for further details Symbol height: 11 mm, with min 5 mm quite zone on both left/ right sides Text below barcode: "WAYBILL" in capital leters, followed by Waybill No. /Shipment ID, 3 mm char height 2D Code - refer to 2D Code specs
Routing Code 22 Symbology Code 128 for ASC MH 10 Application Identifier "2L"; GS1-128 for Application Identifier "403" - - see barcode specs section for further details Symbol height: 11 mm, with min 5 mm quite zone on both left/ right sides The full barcode length must be under 91 mm. Text below barcode: full contents w. DI/AI in parenthesis, w. 3mm char height (9pt)
Customer-owned Information 23 No restrictions other that overall label dimensions. Information (also barcoded) shall not interfere with DHL contents. It may include text, graphics, images.
License Plate (Piece Identifier) Barcode 24 Symbology Code 128 for ASC MH10 License Plates; GS1-128 for SSCC License Plates - see barcode specs section for further details Min quite zones: 5mm (left and right sides). Barcode height must be equal or higher than 25mm, length must be under 91 mm Min x-dim must not be less than 0.33 mm and not higher than 0.51 mm. Text below barcode: full contents w. DI/AI in parenthesis, w. 3mm char height (9pt)

8.2. eWaybill

DHL is required to keep archive copies of all Waybills to meet legal and customer service access requirements. Waybills are usually scanned into an imaging system where the shipment barcode is decoded for indexing purposes. In addition, Waybills for dutiable shipments are scanned into DHL systems (e.g. CIA) to support customs clearance processes.

The eWaybill in this specification has been designed to allow for both Label and eWaybill printing on the same printer. It is, however, also allowed to print the eWaybill on a normal A4 / Letter size document, provided that all specified information is available and the Waybill Barcode is printed.

This Document Type has been designed to support shipments based on DHL's 10-digit Waybill numbering system only. As of 2009, this covers all cross-border Express shipments and all domestic TD shipments.

A 210x110 mm eWaybill can potentially list up to 40 LPs. Where a shipment exceeds this number of pieces, a second page may be printed, to be archived under the same shipment ID.

An eWaybill is either generated through printout and subsequent imaging or as a print file that is archived without being printed.

8.2.1. Size and Orientation

This document type should be **within 95 and 110 mm**. This requirement is met by both A6 labels (portrait) and 4" wide labels. If wider material is used, a frame should be printed around the label content.

There is no hard limit for the document length. The recommended length is **210 mm (A5 width) or 8" (203 mm).** A6 height works as well. A length of 27 cm should not be exceeded, so that proper imaging can be ensured.

8.2.2. Segment Fonts and Sizes

"* eWaybill *" (1) Line 1: char height 4mm (11 pt), bold Line 2: char height 3 mm (9 pt), bold	**Content Code** Char height 6mm (17pt)	**DHL Logo** (2) Min width 30mm. Keep original ratio May be replaced with "DHL" for non-graphics printer
Consignor's Address (3) 7 lines max, min 4. Char height 2 mm (6pt) Postcode, City Name, State/Province, Country (bold font)		**Consignor's Contact Info** (4) All lines 3mm char height (9 pt)
Consignee Address (7) 7 lines max, min 4. Char height 2 mm (6pt) Postcode, City Name, State/Province, Country (bold font)		**Consignee's Contact Details** (9) All lines 3mm char height (9 pt)
(10) **Origin Facility Code** Left-adusted, char height: 5-6 mm (14pt)	**OB Sort Code 7 IB Sort Code** (11) Centered, char height: 4 mm (11 pt)	**Destination Facility Code** (12) Left-adjusted, char height: 5-6 mm (14 pt)
(25a) **Product Info and Payments Terms** Caption " Product Details": 2mm char height (6 pt), bold Line 2 (contents): left-aligned, 3mm char height (9 pt) (25b) Caption "Payer Details": 2mm char height (6 pt), bold All lines (below w. contents): 2mm char height (6 pt)	**"Features / Services"** (26) Caption " Features / Services (Service Code)" : 3mm char height (9 pt), bold, left-aligned All lines, below (contents): 2mm char height (6 pt) All Services, as ordered by customer, to be listed in 4 columns, left-aligned: [Feature] [Service Code] [Service Value] [Currency]	
(16) **Shipment Details** Caption "Shipment Details": 2mm char height (6 pt), bold All lines w. 3mm char height (9 pt) (17b) **Content Descriptions** Caption "Contents": 2 mm char height (6 pt) All Lines with content descriptions: to be printed in 2 columns, with structure [Content Title] [Content Description], 3 mm char height (9 pt) **Regulatory Information (when applicable)** All lines w. 2mm char height (6 pt), bold		
(27) **Shipment Weight and Dimensions (Customer-declared vs DHL-captured)** Contents and respective Captions to be printed in 4 columns: ["Cust Decl Shpt Wgt (UOM)"] ["Dim Wgt (UOM)"] ["DHL SHp Wgt (UOM)"] ["Dim Wgt (UOM)"] Captions: 3mm (9 pt), bold Values: 4 mm (11 pt), in columns, left-aligned		**Piece Count** (19a) Caption: 2 mm char height (6 pt) Value: 4 mm char height (11 pt)
Special Information (if required) (28) All lines w. 2mm char height (6pt)		
Receipt of Customers handing B2C Returns to DHL - Signature Field (29) All captions "Name (in Capital Letters)", "Signature" and "Date (YYYY-MM-DD)" with 2mm char height (6 pt)		
(20) **Shipment Identifier** Symbology: Code 39 - see barcode specs for dimensions allowed; Min quite zones: 5mm (left and right sides). Barcode min height: 11mm Text below barcode: "WAYBILL" in capital leters, followed by Waybill No. /Shipment ID, 3 mm char height		**Special Info** All lines w. 2mm char height (6 pt)
(24a) **Identifiers of Pieces in Shipment** Caption " License Plates of Pieces in Shipment" : 2mm char height (6 pt), bold All lines w. 3mm char height (9pt)	(18a) **Customer Declared Piece Weight** Caption " Cust. Decl. Pc. Wgt (UOM) / Pc. Dim." : 2mm char height (6 pt), bold All lines w. 3mm char height (9pt)	(18b) **DHL Piece Weight** Caption " DHL Pc. Wgt (UOM) / Pc. Dim." : 2mm char height (6 pt), bold All lines w. 3mm char height (9pt)

9. Specifications of Linear Barcodes

9.1. General

The customer can apply more barcodes to the piece, linear as well as 2-dimensional, for his own purposes. However, care must be taken that these other barcodes do not interfere with DHL internal processes. For this reason, the following restrictions apply to other barcodes on the piece:

9.2. Linear Barcode Symbology

DHL uses Code 128 as standard barcode symbology for Piece Identifier and Routing Barcode. For bar-coded information defined under GS1-standards (such as the GS1 SSCC) the GS1-128 barcode symbology has to be used.

For best-possible read-rates of automated scanning, the barcodes on a label should not have the same distance to the left or right edge. This is usually achieved by printing the barcodes centered on the label. Where that is not possible, it is recommended to manually indent the Piece Identifier barcode by at least 2 millimeters.

In accordance with License Plate standards, DHL will also accept Code 39 for ASC MH10 coded information. This barcode symbology however is not the preferred barcode symbology.

CAS *Code 39 must not be used for Piece Identifier barcodes on any labels produced by DHL or printed on systems created by DHL.*

When Code 39 is used by a customer, special care has to be taken that the barcodes do not exceed the maximum length requirements, due to the low information density of this barcode**.**

9.2.1. Code 128 Barcodes

All **BARCODE PARAMETERS** must be as defined in ISO/IEC 16388 “Automatic identification and data capture techniques -- Code 128 Barcode symbology specification”.

Code 128 Barcodes on piece labels and archive labels must meet the following specifications.

Symbol height	The minimum height for the barcodes is 25 mm, unless formally approved (Global SOP) differently. The Routing barcode, may exhibit an exceptional min height of 11 mm, recommended for all Label templates where space is a challenge. Both options are currently available and can be used.
x-dimension	The minimum narrow element dimension (x-dimension) shall not be less than **0.33 mm** and shall also not exceed **0.51 mm** for the ISO/IEC 15417 Standard. The GS1 Standard defines size ranges for the GS1 barcode symbologies, in the EAN General Specifications. For transport packaging, values within the interval module width 0.495 mm to 0.660 mm are allowed. Within the above indicated ranges, the biggest x-dimension should be used that is supported by the printing device used, while complying with barcode length and quiet zone specifications.

Quiet zone	Linear barcode symbols must be printed with leading and trailing **quiet zones. For Code 128, DHL demands not less than 5 mm**. Quiet zones are not part of the barcode itself, i.e. a barcode of 91 mm needs at least 101 mm space on the label.
Quality	The linear barcode quality must, as a minimum, conform to **"Grade B",** tested according to ANSI and CEN (EN 1635), Test Specifications for Barcode Symbols.

9.2.2. Code 39 Barcodes

All **BARCODE PARAMETERS** must be as defined in ISO/IEC 16388 "Automatic identification and data capture techniques -- Code 39 Barcode symbology specification".

Code 39 Barcodes on piece labels and archive labels must meet the following specifications.

Symbol height	The original minimum height for the DHL barcodes is 25 mm. A new barcode version, approved in 2014, has a height of 11 millimeters and is especially recommended for all Label templates where space is a challenge. Both options are currently available and can be used.
x-dimension	The minimum narrow element dimension (x-dimension) must be between 0.36 mm and 0.425 mm.
Code orientation	DHL demands all barcodes to be picket fence-oriented (standing)
Wide-to-Narrow Gap	Between 2.2 and 3.0
Quiet zone	Linear barcode symbols must be printed with leading and trailing quiet zones. For Code 39, DHL demands not less than 6 mm.
Quality	The linear barcode quality must, as a minimum, conform to "Grade B", tested according to ANSI and CEN (EN 1635), Test Specifications for Barcode Symbols.
Start and Stop Char	DHL's Code 39 Waybill barcodes require the standard start- and stop-char "*"
Check Digit	DHL uses the simple MOD 7 check digit: DHL's 9-digit Waybill number is divided by 7, and the remainder forms the check digit, becoming the 10th digit of the Waybill barcode. Example: Waybill number 310003290 • 310003290 divided by 7 is 44286184.29 • Rounded off to next integer: 44286184 • 44286184 * 7 = 310003288 • 310003290 – 310003288 = 2 = Check digit • Barcode content = 3100032902

Symbol height	The minimum height for the barcode is 25 mm. However, a barcode height of at least 28 millimeters is recommended
x-dimension	The minimum narrow element dimension (x-dimension) shall not be less than **0.33 mm**. The x-dimension shall not exceed **0.51 mm**. Within this range, the biggest x-dimension has to be used that is supported by the printing device used, while complying with barcode length and quiet zone specifications.
Quiet zone	Linear barcode symbols must be printed with leading and trailing **quiet zones. For Code 128, DHL demands not less than 5 mm**. Quiet zones are not part of the barcode itself, i.e. a barcode of 91 mm needs at least 101 mm space on the label.
Quality	The linear barcode quality must, as a minimum, conform to **"Grade B",** tested according to ANSI and CEN (EN 1635), Test Specifications for Barcode Symbols.

9.3. Bi-Dimensional Barcode Symbologies

9.3.1. What is a 2D Barcode

A 1D (linear) code is the typical "picket fence" style barcode that people are most familiar with. There are several versions of 1D codes with distinct capabilities: some encode only numbers while others can encode any typical character. Core is that information in the code is organized horizontally from left to right.

2D barcodes are graphical representations (i.e. ordered grid of dark and light dots) that structure, organize and store information both horizontally (as 1-dimensional bar codes) and vertically. They are bordered by a finder pattern. They are also more complex but hold much more information and take up less space than a 1D code. 2D codes require a 2D-enabled (e.g. imager) scanner to be read properly.

The most typical 2D Code Symbologies are:

- Data Matrix
- Aztec Code
- QR Code
- Maxicode
- PDF 417

Currently, the Data Matrix symbology is the most commonly used due to minimal size requirements, robustness by integrated automatic error correction and usability on most 2D devices. PDF417 needs at least 3 times the space than Data Matrix by a reduced reading capability.

9.3.2. Data Matrix

9.3.2.1. Definition

Data Matrix is a two-dimensional matrix symbology made up of square modules arranged in a square or rectangle shaped pattern. This symbology is capable of packing large amounts of data and has the capability of recovering the original data encoded even when it is partially damaged through an error correction technique. Data Matrix codes are characterized by the matrix size which is the overall rows and columns count (for example 32x32 in **Error! Reference source not found.**). Higher matrix sizes need more space and take a higher amount of data.

9.3.2.2. Scope

The Data Matrix symbology will be used to encode operational data to be used in the PUD and In-Facility domains.

9.3.2.3. Key parameters

Within DHL the following code parameters shall be used:

- **Symbology**: Data Matrix ECC 200
- **Module size**: 0.38 mm
  The nominal range of 0.40 > x > 0.38 may, in the future, be extended to the smaller value 0.24 mm due to scanner performance evolution.
  The module size for 2D symbols for Sorting purposes shall not fall below 0.76mm.
- **Symbol matrix**: any symbol matrix is supported.
  As the application typically requires at least 200 characters capacity, only square symbols might be used.
  The smallest possible symbol matrix should be used to get best readability.
- **Macro**: Macro 06 should be used as data is appropriate for that.
- **Quiet Zone**: recommended min 4mm, on all symbol sides
- **Code Quality Grade**: B
- **Printing technology**: any which reaches the quality requirements. Examples: Thermal Transfer, Laser, InkJet etc.
- **Encoding**: Currently, 7 bit data is used only, so no ECI specification required.
- **Human readable interpretation**: The data contents are not shown verbatim on the label. Instead, excerpts like the public information might be displayed similar to the 1D codes.
- **Symbol physical location**: the 2D symbol will be placed, on the label, in such a way as to maximize the quiet zone and protect specially the finder patterns from damage. In addition, It shall meet the ISO 15394:2017 requirements, which specifies the minimum requirements for the design of labels containing two-dimensional symbols on transport units to convey data between trading partners.

9.3.3. QR Code

9.3.3.1. Definition

A QR-Code is a two-dimensional symbology, similar to a Data Matrix or Aztec one, capable of packing large amounts of data and with the capability of recovering the original data encoded even when it is partially damaged, through an error correction technique.

Data encoded in a QR-Code may include alphabetic characters, numbers, double characters and URLs. The symbology uses a small area of square modules with a unique perimeter pattern, which helps the barcode scanner determine cell locations to decode the symbol and extract the data.

Implementation of this symbology is based on the International ISO/IEC Standards 18004:2006 and is also, compliant with the ISO/ IEC 18004:2015 specifications.

9.3.3.2. Scope

The QR symbology will be used to encode operational data to be used in the PUD and In-Facility domains, in the context of the LabelFree scenarios.

9.3.3.3. Key parameters

Within DHL the following reference code parameters [14]should be used:

- **Symbology**: QR Code
- **Module size**:: 0,51mm/20mil
- **Expected (Reference) number of alpha-numeric characters**: 503
- **Symbol matrix version**[15]: 17 => 85 x 85 modules + 4 x 4 modules for the quiet zone.
- **Symbol size:** 44 x 44 mm including quiet zone
- **Error correction level**: H
- **Quiet Zone**: 4*4 modules
- **Code Quality Grade**: C according to ISO/IEC 15415 and ISO/IEC 16480
- **Symbol physical location**: the 2D symbol should be placed on the screen in such a way as to maximize the quiet zone and protect specially the finder patterns from being hidden. This leads to a minimum margin to a screen border of 4 modules.
- **Human readable interpretation**: The data contents are not shown verbatim on the label. Instead, excerpts like the public information might be displayed similar to the 1D codes.

[14] Again, referring to the LabelFree-related business requirements and context

[15] This specific example targets the requirements defined in the context of the LabelFree project

10. Specifications of Material

10.1. Specifications of Paper

Paper for DHL Standard Labels must meet the following specifications:

Paper	
• Color:	Premium white
• Weight :	min. 80 [g/m²]
• Thickness:	90 to 1400 [µm]
	Permanent Adhesive
Adhesive	
Thermal sensitivity (paper, ribbon)	
***** Image density:	1.30 to 1.40 (Macbeth)
* min. activation temperature:	70 C
Resistance:	
• Resistance to: abrasion water	
• Does not degrade when exposed to direct sun	
• Long life (>2 years) if stored in normal conditions	

10.2. Specifications of Label Thermal Adhesive

10.2.1. Description

A paper label with an adhesive backing. It is placed in a thermal printer which uses heat to mark the paper with customer data and barcodes.

10.2.2. Application Example

Customers print data on the label and use the adhesive surface to stick these to the container they are using to ship their commodity.

10.3. Key Technical Data

10.3.1. Specifications

Specification	Value	Units	Tolerance
Glue strength (rolling ball test)	48 (1.8)	mm (in)	±5mm
Thickness of paper	10	µm (in)	±2µm

10.3.2. DHL Standard Specifications List

Standard Specification	
Base unit	**DHL Express Adhesive Label**
Hot melt adhesive	Raflatac RH9 or other adhesive with similar properties
Paper type	Thermal
Minimum temperature for application	0°c
Adhesive operational temperature window	-40°c to +60°c
Recyclable	Yes
Printer	For standard printers up to 200 mm/s

10.3.3. Useful Life

The useful life of label must be 2 years in storage at 20°c. It must be able to travel on the DHL Express network for 10 days with no deterioration in its ability to its adhesion properties.

11. Change Control

This document is controlled by the Network Operations ExCom. Changes or exemptions are not permitted unless submitted and approved as per the Global SOP Express Change and Exemption Request Process.

Any proposal for change must be submitted to the Business Change Request System (BCRS).

CHANGE FROM PREVIOUS RELEASE

CR ID	Change Summary	Section	Current Page Number
tbd	1. Addition of additional segment specifications for the so-called "Highly Compact Label", for both Consignor's and Consignee's Address segments	5.2.1.1, 5.2.4.1, 5.6.1.1, 8.1.6	
	2. Update with new GDPR conditions applicable to both Consignor's and Consignee's Address segments	5.6, 5.8	
	3. More explicit prohibition of Account Numbers (and/ or related information protected through the GDPR Regulation) on Transport Labels	5.12	
	4. Introduction of new syntax for the Label Certification Identifiers (permanent and temporary)	5.2.3	
	5. Relaxation of requirements on location of customer-segment, upon explicit approval through DHL Express' Label Certification Body (now with two options to better meet customer requirements)	5.15	
	6. Relaxation of specifications for the height of the Routing barcode (Amazon CR 2017)	9.2.1	
	7. Addition of a note allowing the relocation of the customer segment after explicit approval from DHL Express	5.13	
	8. General Clean-up and update of Titles and Acronyms to the latest standard (e.g. eWaybill, ESS solutions, License Plate replaced by Piece Identifier – PID, where applicable)	All	
	9. Update of guideline regarding exceptional scenarios, where contact details could be allowed to be printed on the Transport Label, in accordance with the GDPR regulation. Updates were accordingly added to the respective illustrating images.	5.6, 5.8,	
	10. Update of guidelines relative to single (i.e. always matching) "Duty / VAT A/C". No split between both accounts since they always match	6.6.2	
	11. Removal of reference to ACS Label, which is not object of this guideline, from this section	7	
	12. Incoterms removed from Waybill Doc, as per CR 451721	6.6.2	

	13. Reference Loose BBX Transport Label layout, image, updated with individual segments' identification	7.4	
	14. Removal of reference to outdated section (11.1)	5.12.2	
	15. Removal of outdated section 5.11.2 "DHL Shipment Identifier (mandatory)" and corresponding renumbering of the sub-sequent sections	5.11.2	
	16. Original images replaced with higher-quality ones	All sections	
	17. Updated images to reflect latest GDPR guidelines	8	
	18. Piece Identifier length (i.e. number of characters following the Data Identifier) of type JD01 made more clear	5.16.3	
	19. The US Destination Control Statement (US DCS) was removed from the Transport and Waybill Doc images as it is no longer required by the US government. The US government harmonized the US DCS and only requires it to be listed on the commercial invoice.	5.12.1.1	
	20. "Neutral Content Description" section introduced	5.13.1.1	
	21. "Considerations on Highly-Compact Labels" – final layout and segments' dimensional matrix updated	8.1.6	
	22. "Considerations on 2D-enabled Transport Labels" – Phase 2A layout and segments' dimensional matrix updated	8.1.7	
	23. Addition of "2D Data Matrix – Customer Data" section, formally introducing the 2D-enabled Transport Label layout	5.12	
	24. Addition of Bi-Dimensional Barcode Symbologies and symbol attributes	9.3	
	25. "Reverse Video Format" added to Definitions	4.1	
	26. Specification of Medical EXPRESS Logo added	5.10.4	
	27. Correction of the area reserved for the Medical EXPRESS Logo	5.10.4	
	28. Sample image updated to correspond to and reflect specification for 2 decimal points	6.8	

ERRATA

Err ID	Reported by	Date Reported	Current statement says	Recommended to change to	Date Verified	Verifier Name
001	GNOPS Label Team	September.2022	General Typos	Corrected in accordance with Global SOP Dictionary and Glossary of Business Terms	September.2022	GNOPS Label Team

This is an optional section to list changes made to the published document

12. Appendix

12.1. Acronyms and Symbols

Legend for a consistent set of acronyms and symbols throughout the next sections

Acronym	Definition / Meaning
1	Refers to (for example) Segment 1 of reference document type
Char	Relative to characters or to the number of characters of a string
DG	Relative to Dangerous Goods
Doc-type	As per the Global SOP Standard for Classification of Types of Documents
ESS	Electronic Shipping Solutions; formerly known as “eCom”
GS1	Global Standards One – non-profit organization that develops / maintains global standards for business, in this context the well-know GS1-barcodes.
HFC	Relative to Handling Feature Codes
IATA	Relative to the International Air Transport Association
LP	License Plate. Also known as Piece Identifier. The license plate is defined in ISO/IEC 15459-1. It is a unique identifier for transport units. This unique number, regardless of use, shall be specified and assigned by the label issuer and applied to a transport unit and may provide access to traceability data regardless of content and destination. This number shall be **valid for the lifetime of the unit** it is applied to.
Max	maximum
MH10	Accredited Standards Committee MH10 - The ANSI committee for standards for Unit Loads and Transport Packages, including the national standard for bar code labeling
Min	minimum
Opt.	Optional
PID	Piece Identifier
SID	Shipment Identifier

Zip	Generic jargon relative to Postcode, as one possible element of a Postal Location Format